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Infrastructure availability and urban railway performability: A stochastic petri net approach to resilience, congestion, and service reliability

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ABSTRACT

Railway systems are critical infrastructure in urban mobility, yet the reliability of signaling and control systems remains a key determinant of service quality and passenger safety. This study proposes integrated Stochastic Petri Net (SPN) models to evaluate how the availability of Rail Traffic Signaling and Control Systems (RTSCS) impacts railway performability and broader urban outcomes. The framework jointly assesses mean travel time, headway, trips per hour, passenger accumulation per platform and per train, infrastructure availability, and the probability of dual track-circuit (TC) occupancy as a proxy for collision exposure. Using real operational data from the METROREC South Line metropolitan railway in Recife, Brazil, we simulate scenarios with varying train counts and Traffic Flow Routes (TFRs) as redundancy mechanisms. Results show that improving signaling redundancy reduces mean travel time by over 34%, increases service frequency by 52%, and decreases passenger load per train by 34.25%. Track circuit availability improved from 0.856 to 0.959 with six TFRs, while the probability of dual TC occupancy decreased from 0.065% to 0.035%. Enhanced railway availability may also support modal shifts from road transport, contributing to lower CO_2 emissions. By linking infrastructure availability to passenger accumulation, service reliability, and safety, this study provides a quantitative decision-support framework for urban transport planners seeking to improve railway resilience and environmental sustainability.

1. Introduction

As cities pursue net-zero emission targets and seek to reduce dependence on private motor vehicles, reliable rail transport has become a cornerstone of sustainable urban mobility transitions [1,36,37]. Beyond infrastructure expansion, however, the operational reliability of railway signaling and control systems plays a decisive role in determining service quality, passenger accumulation, and overall system resilience [38].

Service unreliability in urban rail networks manifests in increased headway, recurrent delays, and passenger overcrowding. These operational disruptions extend beyond technical inefficiencies: they affect commuter well-being, workplace punctuality, and urban productivity. Empirical evidence has consistently shown that overcrowding and service unpredictability in transit systems are major sources of transport-related stress, with negative consequences for psychological health and life satisfaction [19,35]. Similarly, commuting stress has been linked to impaired in-role job performance and reduced organizational citizenship behavior, operating primarily through the elicitation of negative emotional experiences [41]. Therefore, infrastructure availability is not merely a technical performance indicator; it is a determinant of urban quality of life.

From an environmental perspective, railway reliability also influences modal choice behavior. When rail services are predictable and efficient, they attract riders away from private car use, thereby contributing to reductions in traffic congestion and greenhouse gas emissions [37,39]. Conversely, frequent failures or safety risks may discourage public transport use, reinforcing car dependency and associated environmental impacts. In this sense, the availability of signaling and traffic control systems directly affects broader urban sustainability goals [18].

A smaller body of work has focused specifically on signaling systems as a primary source of failure in urban rail: analyses of operational disruption data from the Beijing metro identify signaling subsystems as one of the leading sources of service disruptions, with cascading effects on headway and passenger loads [38]. Despite this relevance, most

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railway studies focus on either infrastructure reliability or operational performance in isolation. Limited research quantitatively integrates signaling availability, train dynamics, passenger accumulation, and safety risk within a unified analytical framework capable of supporting urban planning decisions. In particular, the relationship between infrastructure redundancy, passenger crowding, service frequency, and collision risk has not been thoroughly examined from a performability perspective [31,34].

To address this gap, this study proposes an integrated Stochastic Petri Net (SPN) framework to evaluate how the availability of Rail Traffic Signaling and Control Systems (RTSCS) affects railway performance indicators and their urban implications. The proposed performability model jointly estimates mean travel time, headway, number of trips per hour, passenger accumulation per platform, passenger load per train, infrastructure availability, and the probability of dual track-circuit (TC) occupancy as an indicator of collision exposure.

The choice of SPN over alternative formalisms is deliberate. Markov chains offer a rigorous basis for availability and reliability analysis, but they require the analyst to enumerate the full state space explicitly, which quickly becomes intractable as the number of trains, track circuits, and concurrent events grows [6]. Bayesian networks are well suited to causal and probabilistic risk inference, yet they do not natively represent concurrency, synchronization, resource contention, or time-dependent stochastic behavior, all of which are central to railway traffic and signaling dynamics [14]. Stochastic Petri Nets, in contrast, compactly capture concurrency, synchronization, and timed stochastic events, scale more gracefully to large concurrent systems, and can still be solved through their underlying Markov chain when an analytical result is required, which makes them particularly appropriate for jointly modeling signaling availability, train movement, passenger accumulation, and collision risk within a single performability framework.

Using real operational data from a metropolitan railway network in Recife, Brazil, the model simulates different operational scenarios, including variations in the number of trains in service and the implementation of Traffic Flow Routes (TFRs) as redundancy mechanisms. By explicitly linking signaling availability to passenger crowding and service reliability, the framework enables an assessment of how technical improvements may contribute to an enhanced commuter experience, reduced operational risk, and improved urban mobility performance.

The main contributions of this research are threefold. First, it integrates availability and performance modeling into a unified performability approach tailored to urban rail systems. Second, it quantifies the systemic effects of redundancy strategies (TFRs) on congestion, safety, and service frequency. Third, it provides a decision-support tool capable of informing infrastructure planning strategies aimed at improving railway resilience and promoting sustainable urban mobility.

2. Related works

This section reviews recent studies, published from 2021 onward, on the performance, availability, and safety evaluation of railway systems, grounded in Petri net formalisms and in complementary modeling approaches. Toguyéni [33] proposed an approach to the critical issue of traffic fluidity in railway systems, aiming to increase traffic efficiency at railway nodes through hardware resources. The approach uses Petri nets to synthesize controllers, modeling the potential routes for different railway traffic through a node and yielding a closed-loop control model that ensures collision prevention.

Han et al. [16] proposed a passenger flow-based model to quantify the resilience of metro stations under operational disturbances. A passenger delay coefficient and a station service capacity metric are coupled to measure how a station preserves its service level during sudden passenger surges, considering passenger flow control and train capacity constraints.

An approach to rescheduling an extensive railway system in the face of disasters is presented in [11], combining a Petri net-based approach with Markov Decision Processes (MDP) to handle the uncertainty of recovery time and a distributed constraint optimization method with autonomous agents to generate the new schedule. Evaluated on real data from East Railways in India, the framework significantly reduces train delays compared to existing approaches.

The authors in [40] highlighted the importance of the Centralized Traffic Control (CTC) system in ensuring the safety and efficiency of rail transport. The article proposes a formal approach using stochastic Colored Petri Nets to quantitatively assess the operational efficiency of CTC with various safety control schemes. The approach includes preventing passenger trains from overtaking freight trains on adjacent tracks. The results of the case study demonstrate the feasibility of this active approach in optimizing real-time emergency management.

Saleh et al. [29] focused on developing an optimized asset management model to reduce operating costs, maintain service quality, and improve safety on rails. The authors propose using Reinforcement Learning (RL) to optimize decision-making processes. In contrast, a Petri net model is suggested as suitable to complement RL, especially in railway maintenance modeling. The study integrates Petri nets with RL to create an effective tool to model and optimize decision-making in the maintenance of the railway section.

Gu et al. [15] developed a vulnerability envelope model for urban rail transit networks subject to simultaneous disruptions of stations and line sections. Using single-level optimization, the authors introduced a route redundancy measure that captures the availability of alternative routes and the service quality experienced by passengers when parts of the network become unavailable.

Milkovic et al. [25] developed a reliability-based model to optimize spare parts inventory and personnel allocation for passenger railway vehicle maintenance. Their study employs regression analysis and neural networks to model the correlation between resource costs and reliability levels, aiming to support strategic fleet management decisions. In contrast, Bianchi et al. [5] conducted a systematic review of Railway Infrastructure (RI) monitoring, examining how the transition from preventive to predictive maintenance affects the reliability and availability of the permanent way.

While the former proposes a quantitative optimization tool for rolling stock, the review by Bianchi et al. emphasizes the importance of structural health monitoring and well-planned maintenance of the superstructure to mitigate critical failures, such as rail breaks, and ensure operational safety. Both studies converge on the conclusion that efficient maintenance management is the fundamental pillar for performability and cost reduction in the railway sector.

Beyond Petri-net-based approaches, other formalisms have also been applied to railway dependability and risk analysis. Chang et al. [6] used a Markov model to compare the reliability of a conventional communication-based train control system and a software-defined architecture with warm-standby redundancy, quantifying the gain in mean time between failures. Feng et al. [14] developed a Bayesian network-based risk evaluation model for railway freight operations to identify and rank operational risk factors. Markov approaches, however, are constrained by state-space growth in highly concurrent systems, and Bayesian networks do not naturally express timed, concurrent dynamics, which motivates the adoption of SPN in this work.

Table 1 summarizes the essential characteristics of the related work discussed. Among the reviewed studies, Zhang [40] and Saleh [29] are the closest to the present scope regarding the classical performance, availability, and safety triad; yet each addresses only two of these three aspects. Passenger crowding and route redundancy have only recently been addressed, and by separate studies: passenger crowding in metro stations [16] and route redundancy in transit networks [15], outside of an integrated dependability analysis.

As the table shows, no prior study combines the three classical aspects with the explicit quantification of passenger crowding, at the platform and train levels, and of the Traffic Flow Route redundancy impact on availability and performance, within a single performability

Table 1
Comparison of the related work according to the railway system considered, the performance and dependability aspects addressed, and the passenger crowding and route redundancy (Traffic Flow Route, TFR) analysis.

Author(s), year	System(s)	Performance and dependability			Demand and redundancy	
		Performance	Availability	Safety	Passenger crowding	Route redundancy
[33]	RI and CTC	✓	–	✓	–	–
[11]	RI	✓	✓	–	–	–
[40]	CTC	✓	–	✓	–	–
[29]	RI	–	✓	✓	–	–
[25]	RV	–	✓	–	–	–
[15]	RI	✓	✓	–	–	✓
[5]	RI	–	✓	–	–	–
[16]	RI	✓	–	–	✓	–
This work	RI and CTC	✓	✓	✓	✓	✓

RI: Railway Infrastructure; CTC: Centralized Traffic Control; RV: Railway Vehicles. The symbol ✓ indicates that the aspect is explicitly modeled or quantified by the study, and the Availability column also encompasses reliability-oriented analysis. Passenger crowding: quantification of passenger crowding per platform and per train; Route redundancy: assessment of the Traffic Flow Route (TFR) redundancy impact on availability and performance.

model that also computes the probability of dual TC occupancy. This integrated scope, rather than the adoption of Stochastic Petri Nets in itself, constitutes the main contribution of this work relative to the reviewed literature.

3. Preliminaries

This section presents the main concepts and definitions regarding the selected railway, the signaling, and the traffic control system used in this railway network.

3.1. Railway and traffic signaling and control system

The RTSCS analyzed in this study is operated by Recife Metro (METROREC), a company subordinate to the Brazilian Urban Train Company (CBTU). The urban rail network in Recife, the capital of Pernambuco state in Brazil, extends for 71 km and comprises three lines: two electrified lines (South and Central) and one diesel-powered line operated with light rail vehicles. The system transports approximately 400,000 passengers per day [8].

The South line, which is adopted in this paper, comprises a stretch of approximately 14 km, consisting of 12 stations. Recife Sul (RECS), Joana

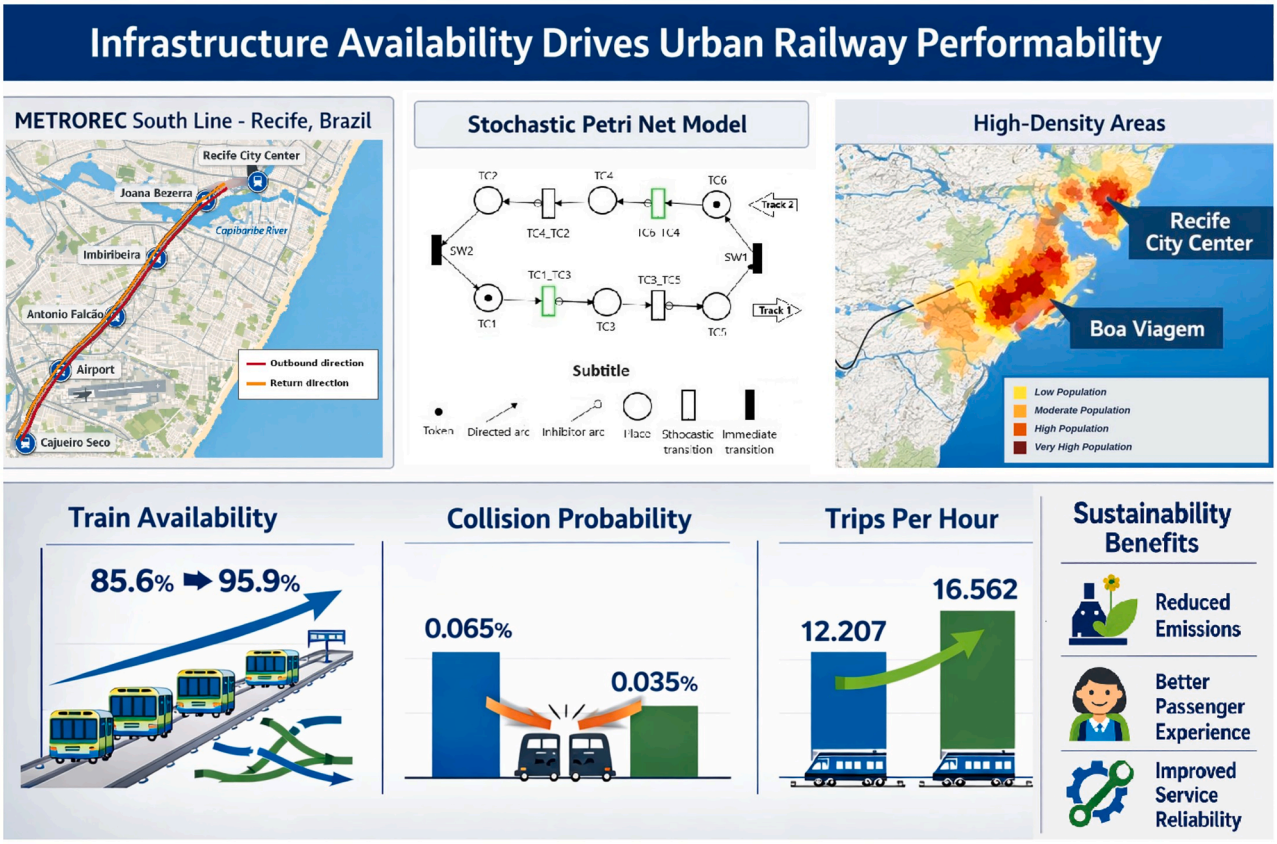


Fig. 1. Graphical Abstract: Infrastructure Availability Drives Urban Railway Performability. The study integrates Stochastic Petri Net modeling with real operational data from METROREC South Line (Recife, Brazil) to evaluate how signaling and traffic control system availability impacts railway performance indicators across high-density demographic areas (Recife City Center: 27,939 inhabitants; Boa Viagem: 125,805 inhabitants). Key results show that infrastructure redundancy mechanisms (Traffic Flow Routes) improve train availability (85.6% to 95.9%), reduce the probability of dual TC occupancy (0.065% to 0.035%), and increase service frequency (12.207 to 16.562 trips/hour), with cascading benefits for passenger experience, urban emissions reduction, and enhanced system resilience.

Bezerra Sul (JOAS), Largo da Paz (PAZ), Imbiribeira (IMB), Antônio Falcão (FAL), Shopping (SHO), Tancredo Neves (NEV), Aeroporto (PTO), Porta Larga (LAR), Monte Guararapes (GUA), Prazeres (PRZ), and Cajueiro Seco (CAJ). The South line serves areas of significant demographic importance in Recife, particularly the city center (Santo Amaro), the commercial district of Boa Viagem, and residential zones in the southern region. Fig. 1

According to the 2022 Brazilian Census (IBGE), Santo Amaro district hosts 27,939 inhabitants and represents the primary employment hub, while Boa Viagem accommodates 125,805 residents and constitutes the second-largest commercial and service center in Recife [17]. The metropolitan region of Recife comprises approximately 4.3 million inhabitants with an overall density of 1327 inhabitants/km^2 [7]. The South line's strategic positioning across high-density demographic areas underscores the critical importance of infrastructure availability for urban mobility and quality of life.

It is worth clarifying that, although the South line is physically an end-to-end route between Recife City Center and Cajueiro Seco, its operation is bidirectional and continuous: each train travels from one terminus to the other along one track, reverses at the terminal station, and returns along the parallel track, repeating this cycle throughout the operating day. As illustrated in Fig. 2, the outbound direction (red) and the return direction (orange) together form a closed operational loop. For this reason, the performance submodel is represented as a cyclic SPN: the cyclic topology does not assume a physically circular line, but rather captures the round-trip behavior of the rolling stock, in which a train that reaches a terminus becomes available again for a trip in the opposite direction. This representation preserves the operational properties of interest for the proposed metrics, namely the number of trips per hour, headway, mean travel time, and passenger accumulation, since these quantities depend on the round-trip cycle time and on the number of trains circulating, rather than on whether the line is geometrically open or closed. The terminal stations are modeled so that a train can only resume operation after completing its dwell and reversal at the end of the line, which ensures that the cyclic model faithfully reproduces the timing of the end-to-end service.

Train traffic on the South, Central, and Diesel lines is managed by an Operational Control Center (OCC). In the case of the South and Central lines, the OCC is equipped with an RTSCS. The RTSCS allows the OCC to monitor the real-time location of all trains on the railway, execute remote commands, and configure train routes. The RTSCS operating on the South line features a series of devices implemented at specific stations (Master or Satellite) and along the railway itself (field equipment). These devices are hierarchically associated, where the failure (unavailability) of one device can affect the others. All data traffic from the RTSCS is controlled by a Communication Manager (CM) and processed by the Train Movement Controller (TMC), which also acts as a multiplexer and connects to field equipment such as signals and Switch machines (SWs).

The TMC also manages the devices responsible for Track Circuit Electronics (TCE). The TCE monitors train occupancy detection and transmits speed codes to trains along all Track Circuits (TC) that integrate the field signaling system [2]. SWs allow trains to change tracks without losing contact between the wheels and the rails. This not only helps in safely and efficiently diverting trains but also increases flexibility in designing the layout of the railway network. Switch machines are the only moving elements of the railway infrastructure [12], since trains, locomotives, light rail vehicles, railcars, and similar vehicles are not part of the fixed infrastructure but belong to the rolling stock.

In addition to transmitting real-time field data to the OCC, the RTSCS informs the Automatic Train Control (ATC) system of a speed code that must be followed during travel. All trains are equipped with an ATC system onboard that communicates with the RTSCS. The train operator must adhere to the speed set by the RTSCS and captured by the ATC; otherwise, the onboard system will apply an emergency brake (safety criterion). Fig. 3 illustrates how 4 SWs are used along a railway network to set up a train's travel route.

In the scenario shown in the figure, the train on track 1 follows the usual route: TC1-TC3-TC5. Each track circuit is separated by insulated joints (end posts) that physically and electrically isolate the track circuits from each other. If TC3 is unavailable due to a failure, the set of A-B switches is activated, allowing the train to move in the reverse direction to TC4 on track 2. Then, the C-D switch set directs the train back to its original path at TC5 on track 1. Therefore, if TC3 fails, the route TC1-TC4-TC5 is used (traffic flow route - TFR). It is important to note that the TFR can only be used when TC4 and TC6 on track 2 are not occupied by any trains, which is a safety measure to prevent train collisions. Fig. 4 illustrates the architecture of the RTSCS on the South line. It should be noted that the traffic signaling and control system is formed by three domain regions, whose scope is delimited by blue lines.

3.2. Performance and availability evaluation with SPN

The following description of the SPN elements has been made more



Fig. 2. Geographic distribution of METROREC South Line stations.

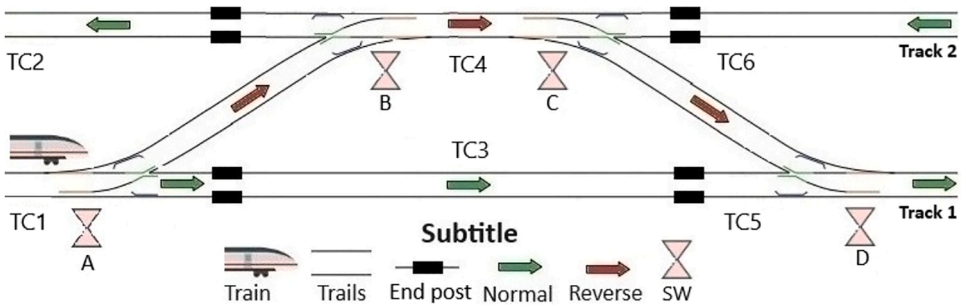


Fig. 3. Functional aspects of a set of SWs forming a TFR.

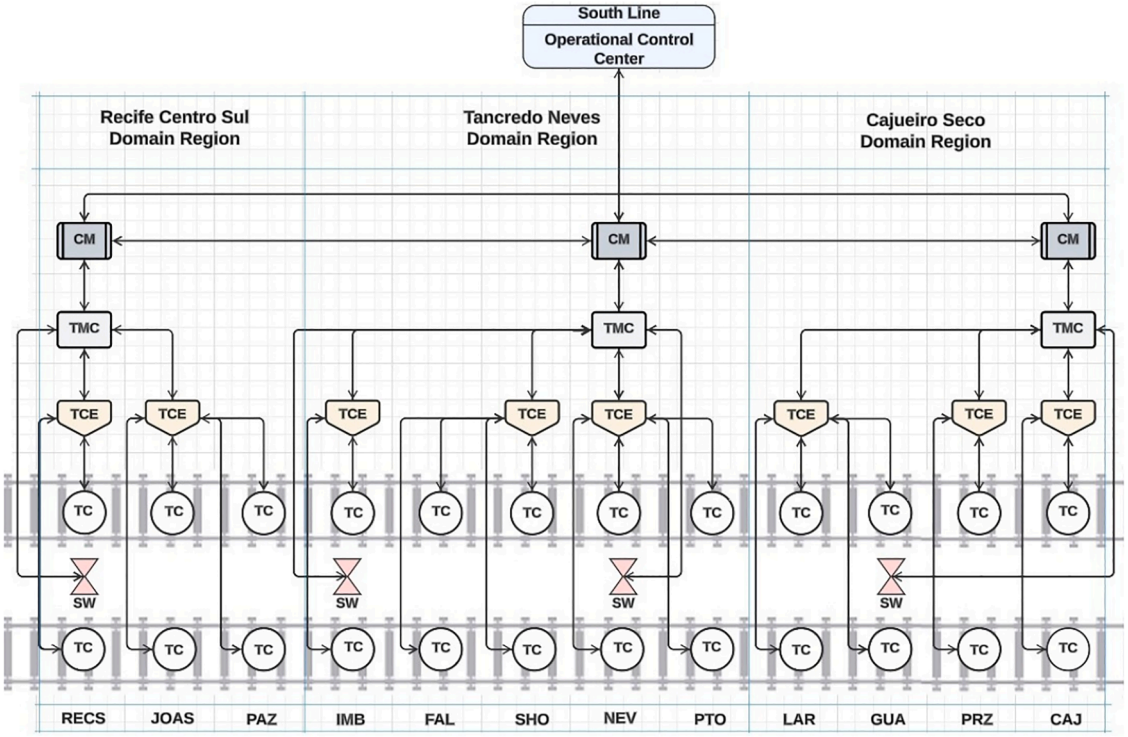


Fig. 4. Architecture of the RTSCS - South line (METROREC).

concise. Petri nets represent a mathematical modeling tool that consists of active elements (transitions) and passive elements (places) connected by directed arcs. SPNs are an extension of Petri nets that allow the inclusion of time associated with transitions originally introduced by Molloy [26] and later generalized by Ajmone Marsan et al. [1], and consolidated in reference texts such as Bause and Kritzinger [4]. Within SPNs, there are two types of transitions: stochastic transitions, which are assigned a time value to represent the duration of an action, and immediate transitions, which take priority over stochastic ones and do not have a time value associated with them. In addition, the inhibitor arc allows a transition to occur when there are no tokens present at the associated place. Places, drawn as circles, represent the local states of the system, such as a track circuit, while transitions represent the events that change these states, such as a train moving to the next track circuit. Tokens, drawn as black dots inside places, carry the dynamic information of the model: the number of tokens in each place defines the current marking (the state of the net), and in this work a token represents a train occupying a track circuit. A transition is enabled when its input places hold the required tokens; on firing, it consumes tokens from its input places and produces tokens in its output places, thereby evolving the net from one marking to the next.

The randomness of the proposed models is introduced through the timed (stochastic) transitions, whose firing delays follow exponential probability distributions, the most widely adopted assumption in stochastic Petri nets [4,23]. Since the exponential distribution has support only on non-negative values, every sampled firing delay is strictly positive, which guarantees that non-positive times are never generated. The parameter of a timed transition represents the mean firing delay rather than a deterministic firing instant, so individual firings may occur earlier or later than this value. When several timed transitions are enabled simultaneously, the firing competition among them, resolved by a race policy, drives the stochastic evolution of the model [34]. As the objective is to evaluate long-run performance and availability measures, the exponential assumption suitably represents the average temporal behavior of the system, and the underlying continuous-time Markov chain provides the analytical solution of the evaluated metrics.

Fig. 5 shows an example of *An spn* representing the movement of 2 trains (tokens) through 6 TCs (places). The stochastic transitions *TC1_TC3, TC3_TC5, TC4_TC2,* and *TC6_TC4* represent the time (Δt) required for a train to move between the track circuits. Therefore, such a Δt is represented as a time interval that is proportional to the speed of the train in each segment. The immediate transitions *SW1* and *SW2* represent the change from one direction to another. Immediate transitions have higher priority over others and can be fired at time zero

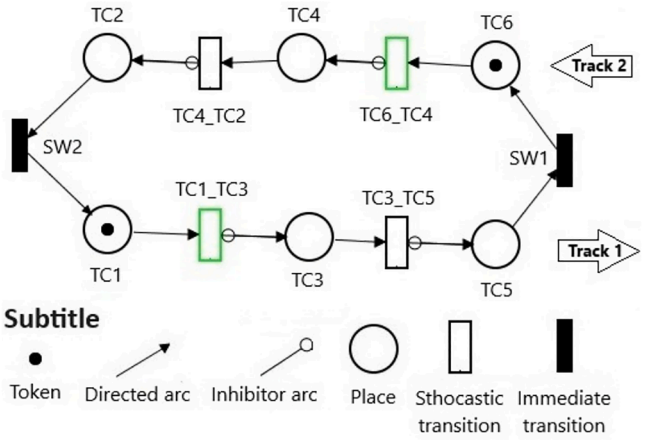


Fig. 5. SPN model that represents the movement of 2 trains on a cyclic rail.

(instantaneously).

The inhibitor arcs present in the stochastic transitions establish that these transitions can only be activated when the subsequent place (track circuit) is free (without a train/token). In the case of the immediate transitions observed in Fig. 5, no associated directed arc is observed. Therefore, a dual occupancy of a track circuit, with the associated risk of train collision, can occur during track merging when there is more than one token at the location following the immediate transition. The reader can notice that the stochastic transitions *TC1_TC3* and *TC6_TC4* are highlighted in green, which indicates that these transitions are able to fire. The firing of the transition *TC1_TC3* will consume a token from the place (*TC1*) and generate a token in the place (*TC3*), simulating the movement of the train from TC 1 to 3.

Using the SPN model shown in Fig. 5, we can define a metric that computes the probability of dual occupancy of a track circuit ($Prob_{DualTC}$). This metric represents the probability (P) of having more than one train (token) in any TC (place) on the rail. A dual occupancy of a track circuit entails a risk of collision between the trains, since unforeseen factors (driver distraction, a sudden health issue, and so on) may prevent the driver from applying the emergency brake in time, even at the restricted speed of 10 km/h adopted in Shunt mode [27]. Eq. (1) presents this metric using the Mercury tool standard [30].

$$Prob_{DualTC} = P\{(\#Plat1 > 1)OR(\#Plat2 > 1)OR(\#Plat3 > 1)OR(\#Plat4 > 1)OR(\#Plat5 > 1)OR(\#Plat6 > 1)\} \quad (1)$$

3.3. Service reliability, crowding, and passenger well-being

Service unreliability in urban rail networks typically manifests as increased headway variability, recurrent delays, and passenger overcrowding. These operational disruptions extend beyond technical inefficiencies and may generate measurable social and economic impacts on daily commuters. Empirical studies demonstrate that crowding in rail environments is associated with elevated physiological stress levels and reduced travel satisfaction [13]. Perceived invasion of personal space and prolonged exposure to high-density conditions may contribute to discomfort, anxiety, and negative commuting experiences [10]. Such effects accumulate over time, potentially influencing overall perceptions of service quality.

From an economic perspective, passenger crowding constitutes a generalized cost of travel and directly affects demand modeling and perceived service attractiveness [32]. Beyond its economic dimension, passenger overcrowding is also a documented safety issue: it has been reported by the Rail Safety and Standards Board (RSSB), for instance in the context of large-scale events such as the London Olympic Games, and the platform train interface is recognized as one of the highest safety-risk areas on the railway, where "a passenger falling from the platform onto the track" is an officially recognized railway safety hazard (Rail Safety and Standards Board [28], 2023). This reinforces that controlling passenger accumulation at platforms is not only a matter of service quality but also of operational safety. In parallel, travel time reliability has been shown to significantly influence user preferences, with passengers often valuing reductions in variability as much as reductions in mean travel time [3,20]. This highlights that headway stability and delay reduction are critical components of perceived service quality.

Consequently, infrastructure availability and redundancy mechanisms may indirectly influence commuter well-being, behavioral responses, and long-term modal choice patterns. Modeling passenger accumulation, headway variability, and signaling performance within a unified performability framework, therefore, enables not only technical evaluation but also an indirect assessment of urban-level consequences associated with service instability.

4. Methodology

This section describes the methodology adopted in this research. The **first step** corresponds to understanding the system, where the logical and operational principles of the RTSCS and the railway are collected and analyzed. Once the system is understood, the **second step** corresponds to the development of the proposed model and the evaluation of the performance and availability metrics. The SPN models represent the failure and recovery of the RTSCS equipment (availability) and the displacement of the train through the track circuits (performance). Examples of metrics that can be computed include the mean travel time, the interval between trains (Headway), the number of trips made per hour, and the availability of each RTSCS track circuit.

The next phase (**third step**) is to validate the SPN model and evaluation metrics through the process of comparing the results obtained by stationary simulation with those expected from mathematical calculation. It should be noted that this numerical validation was complemented by a qualitative validation with system experts: the proposed model was presented to the technicians responsible for restoring the communication and control systems and to the traffic controllers who operate at the operational control center, whose feedback supported the realism of the modeled behavior and helped to ensure that no critical hazard relevant to the analyzed scope was overlooked. In its current form, the model computes the impact of the signaling and control devices on the performance and availability metrics of the system and also calculates the probability of dual TC occupancy, as previously explained. Other hazards regarded as critical can likewise be modeled, requiring only the appropriate adjustments that specifically reproduce the hazard of interest, in the same way as adopted for the collision risk, for which it was established in the model that two trains (tokens) cannot occupy the same track circuit (TC).

Once the validation is satisfactory, the **fourth step** involves simulating different scenarios (e.g., variations in the number of trains in operation). Once these simulations are completed, the phase of proposing improvements to the system is conducted (**fifth step**). In this stage, the implementation of TFRs along the railway is proposed. These TFRs aim to improve the traffic flow on the railway, with a focus on increasing availability.

Finally, the last phase (**sixth step**) is to evaluate the results achieved. The results presented in Section 7 demonstrate that the proposed models enable the generation of data that can be used for train traffic management and control. For example, the impact of introducing each train into the railway can be captured by the model. Furthermore, the optimization achieved through the implementation of the TFRs is demonstrated by the improvements in the results of the evaluated metrics.

5. SPN models and metrics

This section presents the proposed SPN model to assess the impact of

the availability of RTSCS on the performance indicators of the railway. The model consists of three submodels: (i) the availability of RTSCS equipment, (ii) railway performance, and (iii) passenger boarding on the platforms. The following sections discuss each mentioned submodel and their respective evaluation metrics.

5.1. Availability submodels

The availability submodels consist of SPNs that can represent the operational/failure states of the devices in the rail traffic signaling and control system. Fig. 6 shows the availability submodels of each CAJ-DR RTSCS device concerning the hierarchy: most relevant (CM) to least relevant (TCs).

The RTSCS will only be unavailable when all track circuits (TCs) fail simultaneously. This is because each track circuit has its own command line, meaning that the failure of one device may or may not affect the others. Thus, as long as there is at least one operational track circuit, the RTSCS will remain available in the region to which the TC belongs. The architecture illustrated in Fig. 6 represents the hierarchy of devices within the RTSCS structure, where the most critical equipment is always above the others. Thus, a failure in the "TCE_CAJ", for example, will only imply the unavailability of the CAJ1 and CAJ2 TCs. A failure in the TMC leads to the unavailability of all devices except the CM, as the latter occupies a higher position in the system's hierarchical architecture than the train movement controller.

Table 2 presents the parameters used to configure the stochastic transitions of the previous model presented. For each component, the table shows the observation period, cumulative operating hours, number of recorded failure events (*N*), mean *MTTF* and *MTTR* along with their standard deviations (*SD*), the corresponding coefficients of variation (*CV*), the ratio between the mean and median *MTTR*, and the resulting component availability.

The MTTF and MTTR associated with the failure and recovery transitions were computed from the analysis of the historical failure and maintenance records kept in the maintenance-management software (a proprietary system) operated by METROREC for the track circuits (TCs) and the associated devices located in the section of the system that was modeled. For each device, the MTTF was estimated as the mean operating time between consecutive recorded failures, and the MTTR as the mean time elapsed between the registration of a failure and the restoration of the device to operation, both obtained directly from the corrective-maintenance logs of the modeled section. Because these values come from the operator's field records rather than expert estimates, they reflect the actual reliability and maintainability behavior of the installed equipment, supporting confidence in the input parameters used in the availability submodels.

In addition, we processed METROREC's Protheus CMMS corrective maintenance data to compute the sample size and observation window used for each component. After analyzing the data, we obtained 152,960 corrective work order records, which represent our sample size for the period from 2006– 01– 01 to 2026– 01– 01. In addition, the total number of failure/repair events obtained was 3428, which can be obtained by summing column *N* of the table. The number of failures achieved by each component ranges from 12 events for the CM, the most safety-critical and least frequently replaced device, up to 560 for CAJ2, the most numerous and least reliable track circuit, consistent with the reliability hierarchy of Fig. 6. The coefficient of variation ($CV = SD/$ mean) remains at or below 0.25 for both *MTTF* and *MTTR* across all 13

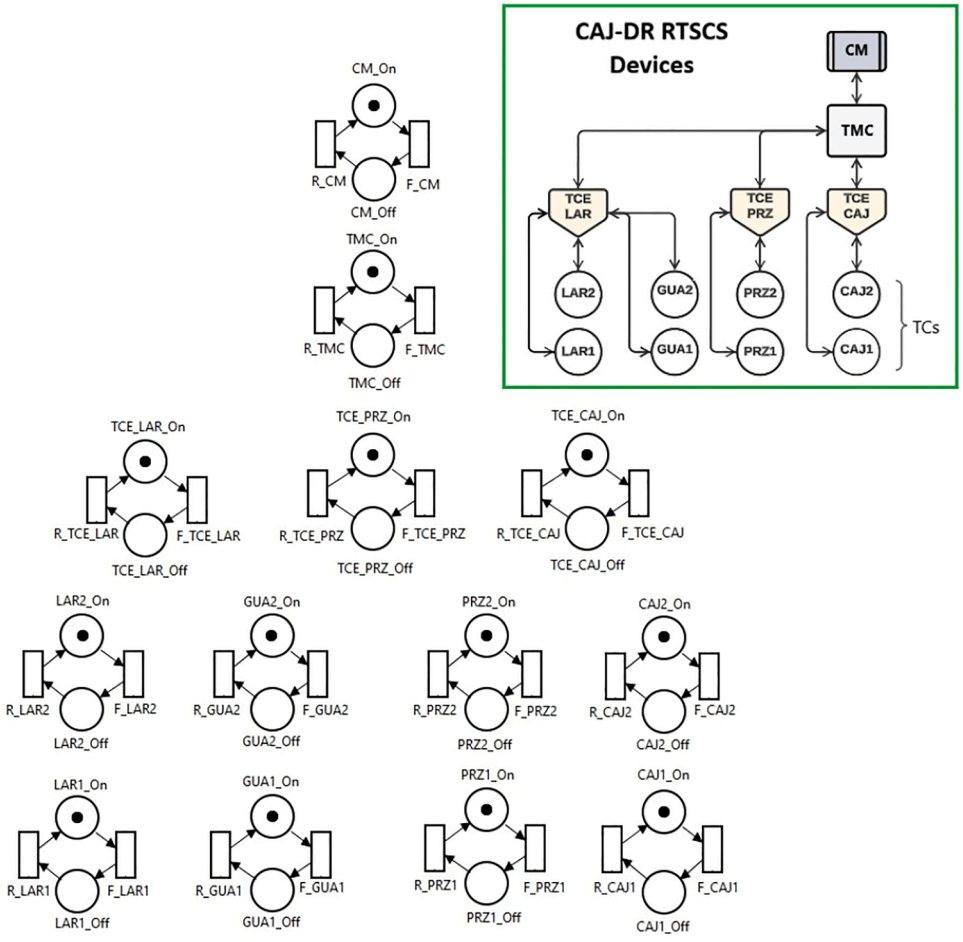


Fig. 6. RTSCS device availability submodels.

Table 2
MTTF and MTTR values inserted into the stochastic transitions of the RTSCS device availability submodels, together with the number of recorded failure events (N), observation period, cumulative operating hours, standard deviations (SD), coefficients of variation, MTTR mean-to-median ratio, and resulting availability computed from the corrective-maintenance records of each device.

Component	Op. hours	N	MTTF mean (SD)	CV(MTTF)	MTTR mean (SD)	CV(MTTR)	MTTR mean/med.	Avail. (%)
CM	144,552.0	12	788,400.0 (184,383.3)	0.23	789.1 (187.4)	0.24	0.94	99.9
TMC	124,841.2	20	394,200.0 (89,800.2)	0.23	790.0 (161.3)	0.20	1.01	99.8
TCE_LAR	118,276.0	28	262,800.0 (53,566.8)	0.20	790.8 (196.5)	0.25	0.97	99.7
TCE_PRZ	116,394.1	32	225,257.2 (48,092.4)	0.21	904.6 (189.5)	0.21	0.98	99.6
TCE_CAJ	114,984.0	36	197,100.0 (39,766.1)	0.20	593.1 (125.6)	0.21	0.99	99.7
LAR1	106,536.8	240	26,725.4 (5133.3)	0.19	4386.8 (1025.0)	0.23	1.02	85.9
LAR2	106,251.0	300	21,308.1 (4204.2)	0.20	3382.7 (781.1)	0.23	1.01	86.3
GUA1	106,070.7	360	17,716.9 (3533.1)	0.20	2836.4 (641.6)	0.23	1.01	86.2
GUA2	105,958.4	400	15,927.3 (3207.5)	0.20	2507.0 (608.2)	0.24	1.01	86.4
PRZ1	105,884.1	440	14,466.1 (2871.2)	0.20	2355.0 (535.5)	0.23	0.97	86.0
PRZ2	105,809.0	480	13,250.4 (2577.7)	0.19	2139.1 (485.8)	0.23	0.98	86.1
CAJ1	105,751.1	520	12,223.3 (2398.1)	0.20	1924.0 (451.0)	0.23	1.00	86.4
CAJ2	105,718.2	560	11,343.9 (2310.4)	0.20	1877.4 (430.2)	0.23	1.01	85.8

components, and the *MTTR* mean-to-median ratio stays close to 1 (0.94 - 1.02), indicating that the underlying inter-failure and repair-time distributions are reasonably symmetric and not dominated by a few extreme observations. This supports using the mean values as representative *MTTF/MTTR* inputs for the stochastic transitions in the availability submodels.

Additionally, the results shown in Table 2 reveal a clear distinction between the central/transmission components (CM, TMC, and TCEs) and the line components. The former exhibit substantially higher *MTTF* values and shorter repair times, resulting in component availabilities above 99.6%. In contrast, the line components present considerably lower *MTTF* values and longer *MTTR*s, leading to availabilities of approximately 86%. The coefficients of variation for both *MTTF* and *MTTR* remain relatively consistent across components, generally ranging from 0.19 to 0.25, indicating comparable dispersion in failure and repair times. In addition, the mean-to-median *MTTR* ratios are close to unity for all devices, suggesting that the mean repair times are representative of the central tendency of the observed repair-time distributions. These results are consistent with railway signaling asset-management studies that likewise rely on multi-year corrective-maintenance logs to characterize the failure and repair behavior of individual field components [21].

5.1.1. Availability metrics

As mentioned in the previous subsection, the RTSCS will only become unavailable when all track circuits fail simultaneously. Analyzing Fig. 6, it is evident that the RTSCS comprises eight track circuits, each associated with its respective device chain. For example, for the LAR1 track circuit to remain operational, the devices CM CAJ, TMC CAJ, TCE LAR, and TC LAR1 must all be in functional condition. Since the availability submodels define the operational equipment as "On=1″ and the failed equipment as "Off=1", the availability of the LAR1 station track circuit ($Avail_{LAR1}$) can be expressed by Metric 2.

$$\begin{aligned} Avail_{LAR1} &= P\{(\#LAR1_On = 1)AND(\#TCE_LAR_On \\ &= 1)AND(\#TMC_On = 1)AND(\#CM_On = 1)\} \end{aligned} \quad (2)$$

The mean availability of other track circuits ($Mean_{Availability(TCs)}$) follows the same methodology, requiring only the necessary adjustments to represent the devices involved in each command chain. Therefore, the mean availability of the track circuits is determined as follows:

$$Mean_{Availability(TCs)} = \frac{1}{n}\sum_{i=1}^{n} Availability_{TCs}(i) \quad (3)$$

Although the arithmetic mean is an elementary operation, Eq. (3) is included to make explicit how the availability of the modeled section is obtained: it is not a generic average, but the mean over the individual availabilities of the n track circuits (TCs) that compose the modeled region of the railway network. The availability of each TC was computed separately from its own availability submodel, and only then combined through Eq. (3) to obtain the mean availability of the section. This quantity is relevant because the performance of the system is influenced by the mean availability of all the TCs that compose it, so making this aggregation explicit clarifies how the availability and performance submodels are linked.

5.2. Performance submodel

The performance model represents the movement of trains (tokens) through the CAJ-DR track circuits (places). Additionally, the model evaluates the movement of trains under two conditions: (i) RTSCS mode, in which the train moves at the speed established by the train signaling and control system; (ii) Shunt mode, which occurs only in the case of a failure that affects the availability of one or more TCs. Fig. 7 illustrates the railway performance submodel.

The necessary requirements to satisfy a movement condition (RTSCS or Shunt) are defined through guard expressions. Moreover, for each track circuit, there will be a stochastic transition with the movement time corresponding to the condition it represents. In other words, the stochastic transitions modeled ahead of the places (stations) are fed with the values of the time interval spent by the train to traverse the corresponding segment; these values are obtained from Table 3 and result from an average-speed calculation, since the inter-station distances and the train speeds are provided by the company that operates the system. The condition that each transition represents refers to whether the train travels in Shunt or in RTSCS mode, since the train may run under one of these two conditions depending on the availability of the signaling and control devices for each segment between stations.

Table 3 compiles the data for all stochastic transitions in the performance submodel.

The movement times listed in this table were obtained from METROREC operational data, namely the section lengths and the operating speeds practiced on each segment of the modeled route, so that each Δt corresponds to the average time a train takes to traverse the respective track circuit.

It is worth noting that the guard expressions presented in Table 3 not only define the triggering criteria for each transition, specifying whether the train operates in RTSCS or Shunt mode, but also establish the connection between the performance submodel and the availability submodels. It can be seen that for each "RTSCS transition", the guard expression defines which devices must be operational ("On=1") so that the TC can be available (RTSCS mode). Thus, for the LAR1 track circuit to be operational, all devices must be available (#LAR1_On=1)AND (#TCE_LAR_On=1)AND(#TMC_On=1)AND(#CM_On=1), as can be seen in the first line of Table 3.

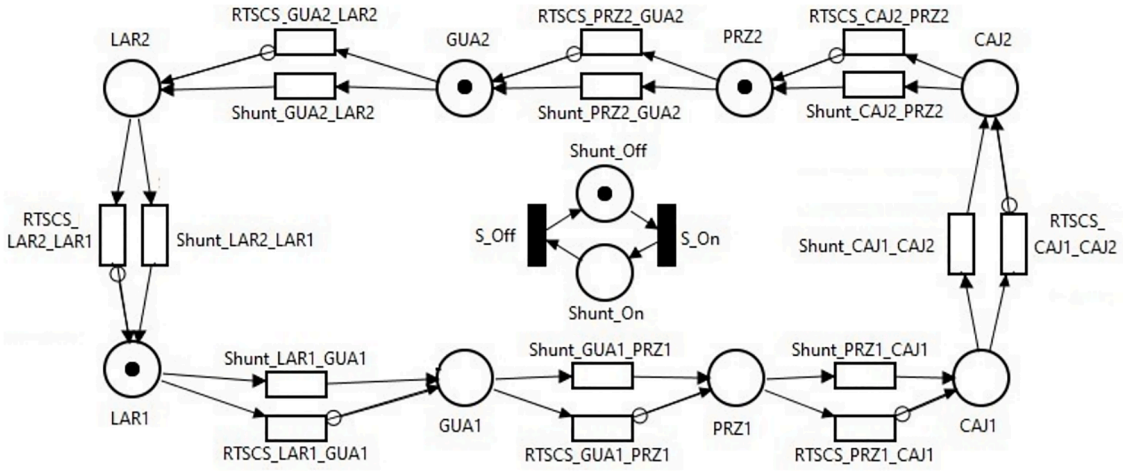


Fig. 7. Performance submodel - 3 trains are observed in operation, occupying the LAR1, GUA2, and PRZ2 track circuits.

Table 3
Stochastic transitions data - performance model.

Transition	Time (min)	Guard expression
RTSCS_LAR1_GUA1	1.247	(#LAR1_On=1)AND(#TCE_LAR_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_LAR1_GUA1	5.236	((#LAR1_Off=1)OR(#TCE_LAR_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_LAR2_LAR1	1.200	(#LAR2_On=1)AND(#TCE_LAR_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_LAR2_LAR1	1.200	((#LAR2_Off=1)OR(#TCE_LAR_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_GUA1_PRZ1	1.45	(#GUA1_On=1)AND(#TCE_LAR_On=1)AND (#TMC_On=1)AND(#CM_On=1)
RTSCS_GUA1_PRZ1	6.234	((#GUA1_Off=1)OR(#TCE_LAR_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_GUA2_LAR2	1.247	(#GUA2_On=1)AND(#TCE_LAR_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_GUA2_LAR2	5.236	((#GUA2_Off=1)OR(#TCE_LAR_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_PRZ1_CAJ1	1.553	(#PRZ1_On=1)AND(#TCE_PRZ_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_PRZ1_CAJ1	6.831	((#PRZ1_Off=1)OR(#TCE_PRZ_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_PRZ2_GUA2	1.417	(#PRZ2_On=1)AND(#TCE_PRZ_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_PRZ2_GUA2	6.234	((#PRZ2_Off=1)OR(#TCE_PRZ_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_CAJ1_CAJ2	1.200	(#CAJ1_On=1)AND(#TCE_CAJ_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_CAJ1_CAJ2	1.200	((#CAJ1_Off=1)OR(#TCE_CAJ_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)
RTSCS_CAJ2_PRZ2	1.453	(#CAJ2_On=1)AND(#TCE_CAJ_On=1)AND (#TMC_On=1)AND(#CM_On=1)
Shunt_CAJ2_PRZ2	6.831	((#CAJ2_Off=1)OR(#TCE_CAJ_Off=1)OR (#TMC_Off=1)OR(#CM_Off=1)) AND (#Shunt_On=1)

The guards of the "Shunt transitions" indicate that the firing condition of these transitions requires the failure of any RTSCS device ("Off=1″) and the activation of the Shunt mode (Shunt_On=1 - Fig. 7). Thus, for the train to move in Shunt mode over the LAR1 track circuit (triggering the "Shunt_LAR1_GUA1" transition - Fig. 7), the guard expression is ((#LAR1_Off=1)OR(#TCE_LAR_Off=1)OR(#TMC_Off=1) OR(#CM_Off=1))AND(#Shunt_On=1), as presented in the second line of Table 3.

It should be noted that in Shunt mode, the train is not subordinated to the RTSCS. Consequently, its speed is restricted to 10 km/h, and the operational control center loses the ability to identify the train on the faulty track circuit. Furthermore, ATC and safety criteria are inhibited when Shunt mode is active. Additionally, as observed in Fig. 7, no inhibitor arcs are modeled for the "Shunt transitions". Thus, the train can advance to the next track circuit even if it is occupied. This characteristic implies that the Shunt mode may lead to collisions between trains, particularly if the train driver fails to apply the emergency brake after noticing another train ahead. When the most critical components fail ((#CM_Off = 1) OR (#TMC_Off = 1)), the Shunt mode may be activated regardless of the train's location (i.e., triggering the immediate transition "S_On" in the performance submodel). In the case of a TCE failure, however, Shunt activation depends on the train's position. The conditions for activating Shunt mode ("S_On") are presented in the guard expressions below:

(((#CM_Off=1)OR(#TMC_Off=1))OR(((#TCE_LAR_Off=1)AND ((#LAR1=1)OR(#LAR2=1)OR(#GUA1=1)OR(#GUA2=1)))OR ((#TCE_PRZ_Off=1)AND((#PRZ1=1)OR(#PRZ2=1)))OR ((#TCE_CAJ_Off=1)AND((#CAJ1=1)OR(#CAJ2=1))))OR (((#LAR1=1)AND(#LAR1_Off=1))OR((#LAR2=1)AND (#LAR2_Off=1))OR((#GUA1=1)AND(#GUA1_Off=1))OR ((#GUA2=1)AND(#GUA2_Off=1))OR((#PRZ1=1)AND (#PRZ1_Off=1))OR((#PRZ2=1)AND(#PRZ2_Off=1))OR ((#CAJ1=1)AND(#CAJ1_Off=1))OR((#CAJ2=1)AND (#CAJ2_Off=1)))

The guard expression representing the conditions for deactivating Shunt mode (S_Off) is given by the negation of the activation expression (NOT "S_On"). Finally, in the event of a track circuit failure, Shunt mode will be activated only if the train is occupying the failed track circuit. Thus, the unavailability of track circuit 1 at Porta Larga station will cause Shunt mode activation only if the train is occupying this specific track circuit while the failure occurs. Accordingly: ((#LAR1 = 1) AND (#LAR1_Off = 1)). The deactivation of Shunt mode (i.e., triggering the immediate transition "S_Off") can be represented by the negation of the activation guard condition.

When the firing condition of a transition (RTSCS or Shunt) is satisfied, the token of a place will be "consumed" from the place preceding the activated transition and generated in the subsequent place, simulating the movement of the train (token) between the TCs (places) in the performance submodel. It is also observed that the transitions initiated by "Shunt" do not have an inhibitor arc associated. Therefore, the train can proceed to the next TC in the event of a failure, even if that track

circuit is already occupied.

In the scenario depicted in Fig. 7, for example, a train is on the PRZ2 track circuit (#PRZ2=1), and another is further ahead (#GUA2=1). Thus, in the event of a failure that affects the availability of the track circuit PRZ2, only the "Shunt_PRZ2_GUA2″ transition can be triggered, consuming the token from the "PRZ2″ place and generating a token in the "GUA2" place (#GUA2=2), thus characterizing a dual occupancy of the track circuit, with the associated risk of collision between the trains.

Note that, under the signaling and control system, there is no risk of collision between the trains, since the inhibitor arc modeled in the transitions initiated by "RTSCS" prevents the train (token) from advancing to the next track circuit (place) if it is occupied by another train.

5.2.1. Performance metrics

The metrics generated to evaluate the performance indicators selected in this research are described and listed below.

Number of trains

This metric calculates the expected number of trains (tokens) on any track circuit (place) within the railway (performance submodel). Thus, the number of trains operating ($N^{\circ}Trains$) in the CAJ-DR can be expressed as Eq. (4).

$$\begin{aligned} N^{\circ}Trains &= (E\{\#LAR1\}) + (E\{\#GUA1\}) + (E\{\#PRZ1\}) \\ &+ (E\{\#CAJ1\}) + (E\{\#CAJ2\}) + (E\{\#PRZ2\}) \\ &+ (E\{\#GUA2\}) + (E\{\#LAR2\}) \end{aligned} \tag{4}$$

Mean travel time

Corresponds to the mean time required for a train (token) to travel through all the track circuits (places) of the railway (performance submodel). This metric is based on Little's law [22], which is defined as shown in Eq. (5). It establishes a simple mathematical relation, where "L" represents the mean number of items in the system (trains on the railway), "λ" denotes the mean rate at which trains (tokens) arrive in the system, and "W" corresponds to the mean time a train spends in the system.

It is important to justify the applicability of Little's law in this context. Little's law is a general and distribution-independent result that holds for any stable system operating in a stationary regime, relating the long-run average number of items in the system, the long-run average arrival (throughput) rate, and the long-run average time an item spends in the system, regardless of the arrival or service-time distributions and of the queueing discipline [22]. Its application therefore does not require dynamic signalling; it only requires that the system be stable, that is, that the average input rate equals the average output rate so that the mean number of items remains finite, and that the measures be taken in steady state. In the proposed SPN models, these conditions are satisfied: the net is bounded (the number of tokens, i.e., trains, is conserved and finite), and its evolution is described by an underlying ergodic continuous-time Markov chain whose stationary distribution provides the steady-state expectations used in the metrics [4,23,34]. Consequently, Little's law is applied here at the level of each track circuit (place): for a given place, the throughput is obtained from the stationary expected number of tokens and the mean sojourn time, and the mean travel time of the train is recovered by summing the contributions of the places along its route. This per-place application of Little's law to stationary measures obtained from the underlying Markov chain is a well-established practice in stochastic Petri net performance evaluation [4,23], which ensures that the relationship is valid for the end-to-end service modeled in this work.

$$L = \lambda \times W \tag{5}$$

Thus, the travel time ($Travel Time$ or "W") will be the ratio of the mean number of trains in the system ($N^{\circ}Trains$ or "L") divided by throughput ("λ") as shown in Eq. (6).

$$Traveltime = \frac{N^{\circ}Trains}{\lambda} \tag{6}$$

In SPN, the expected value of a Place (E{#Place}) refers to the mean number of tokens in that place. Thus, track circuit throughput (λ_{TC}) can be represented by the fraction of the mean number of items in a given place (E{#Place}) to the mean time (Δt_{mean}) that this item (token) spends in the place (track circuit).

$$\lambda_{TC} = \frac{(E\{\#Place\})}{\Delta t_{mean}}$$

Thus, considering the LAR1 track circuit, the travel time LAR1 ($Travel_{LAR1}$) is:

$$Travel_{LAR1} = \frac{N^{\circ}Trains}{\lambda_{LAR1}} = \frac{N^{\circ}Trains}{\frac{(E\{\#LAR1\})}{\Delta t_{LAR1}}}$$

The mean time associated with LAR1 (Δt_{LAR1}) is defined by the availability of TC LAR1, since it considers the time during which the train travels under full operational conditions (RTSCS available) and the time during which the train travels in Shunt mode (RTSCS unavailable). Given this information and the time values for each condition (RTSCS and Shunt) presented in Table 3, the mean time associated with the LAR1 track circuit is defined as Eq. (7):

$$\begin{aligned} \Delta t_{LAR1} = [(P\{(\#LAR1_On = 1) AND (\#TCE_LAR_On = 1) AND (\#TMC_On = 1) \\ AND (\#CM_On = 1)\}) \times \Delta t_{RTSCS_LAR1}] + [(1 - P\{(\#LAR1_On = 1) \\ AND (\#TCE_LAR_On = 1) AND (\#TMC_On = 1) AND (\#CM_On = 1)\}) \times \Delta t_{Shunt_LAR1}] \end{aligned} \tag{7}$$

Note that the first part of the metric, up to the addition operator, refers to the time interval in which the train travels in RTSCS mode (P {(#LAR1_On = 1) AND (#TCE_LAR_On = 1) AND (#TMC_On = 1) AND (#CM_On = 1)}) × 1.247). The second part, after the addition, refers to the time interval in which the train travels in Shunt mode ((1 − P {(#LAR1_On = 1) AND (#TCE_LAR_On = 1) AND (#TMC_On = 1) AND (#CM_On = 1)}) × 5.236).

The same methodology adopted to define the mean time associated with TC LAR1 applies to the other track circuits in the system. Thus, after considering the throughput of all TCs, the mean travel time can be calculated based on the track circuits of the system ($MeanTravel_time$). Thus:

$$\begin{aligned} MeanTravel_time = \frac{1}{8}\Bigg(&\frac{N^{\circ}Trains}{\lambda_{LAR1}} + \frac{N^{\circ}Trains}{\lambda_{LAR2}} + \frac{N^{\circ}Trains}{\lambda_{GUA1}} \\ &+ \frac{N^{\circ}Trains}{\lambda_{GUA2}}\Bigg)\Bigg(\frac{N^{\circ}Trains}{\lambda_{PRZ1}} + \frac{N^{\circ}Trains}{\lambda_{PRZ2}} \\ &+ \frac{N^{\circ}Trains}{\lambda_{CAJ1}} + \frac{N^{\circ}Trains}{\lambda_{CAJ2}}\Bigg) \end{aligned} \tag{8}$$

In the equation above, there is a value in parentheses (from 1 to 8) that refers to the index "i" of the term to which it pertains, allowing the representation of the mean travel time using summation notation. Thus:

$$MeanTravel_time = \frac{1}{n}\left[\sum_{i=1}^{n}\left(\frac{N^{\circ}Trains}{\lambda_{TC}}\right)(i)\right] \quad (9)$$

Headway

Calculates the time between the passage of one train and the next on any track circuit of the railway. This metric is the result of dividing the mean travel time by the number of trains operating.

$$Headway = \left(\frac{MeanTravel_time}{N^{\circ}Trains}\right) \quad (10)$$

Number of trips made per hour

The number of trips made per hour ($N^{\circ}Trips/hour$) considers the number of trips a train makes in a time interval of 1 hour (60 min) and multiplies this value by the number of trains in operation ($N^{\circ}Trains$).

$$N^{\circ}Trips\Big/hour = \left[\left(\frac{60}{MeanTravel_time}\right) \times N^{\circ}Trains\right] \quad (11)$$

Probability of Dual TC Occupancy

Calculated by the probability (P) of having more than one train (token) on the same track circuit (place). Therefore, since two trains cannot occupy the same TC at the same time, a value higher than one token at a place that represents a TC indicates a dual occupancy of the track circuit. A dual occupancy of a track circuit entails a risk of collision between the trains, since unforeseen factors (driver distraction, a sudden health issue, and so on) may prevent the driver from applying the emergency brake in time, even at the restricted speed of 10 km/h adopted in Shunt mode. For example, considering the LAR1 track circuit, the probability of dual TC occupancy is given by: P(#LAR1>1). It is worth noting that this metric does not consider the possibility of the driver's reaction (emergency brake) in the event of an imminent collision between trains. In this way, the probability of dual TC occupancy in the CAJ-DR is:

$$\begin{aligned} Prob.DualTC = P\{&(\#LAR1 > 1)OR(\#LAR2 > 1)OR(\#GUA1 \\ &> 1)OR(\#GUA2 > 1)OR(\#PRZ1 > 1)OR(\#PRZ2 \\ &> 1)OR(\#CAJ1 > 1)OR(\#CAJ2 > 1)\} \end{aligned} \quad (12)$$

Regarding the safety interpretation of this metric, it should be clarified that the model adopts the assumption that only one train can occupy a track circuit at a time. Consequently, an occupancy of a track circuit by two trains (tokens) characterizes a collision strictly within the criteria and requirements considered in the modeling. For one train to overtake another, for instance, the leading train would have to wait at a track circuit while the following train diverts to the parallel track (the track circuit parallel to the one occupied by the leading train) and, after passing it, returns to the original track; any configuration in which two trains share the same track circuit is, by the modeling criteria, counted as a collision. It is also acknowledged that, at the Shunt-mode speed of 10 km/h, the driver could indeed apply the emergency brake; however, as already discussed for the dual-TC-occupancy metric, the criteria considered for this collision scenario deliberately do not account for the driver's reaction, since the objective is precisely to compute the probability of collision in the absence of such a reaction, whether due to non-observance, an unforeseen event, a lapse of attention, a sudden health issue, and so on.

Number of passengers per platform

Estimates the mean number of passengers waiting on a platform before boarding a train. It is defined based on two parameters:

1. the passenger arrival rate on the platform, denoted by $\lambda_{platform}$ (passengers per unit of time), and
2. the train Headway, denoted by *Headway* (time interval between consecutive trains).

Let the arrival time of passengers on a platform be represented by:

$$T_{platform} = \frac{1}{\lambda_{platform}}$$

where $T_{platform}$ is the mean time between the arrival of two consecutive passengers.

During the interval *Headway*, the expected number of passengers arriving on the platform is proportional to the arrival rate. Therefore, the expected number of passengers waiting on a platform is given by:

$$N_{platform} = \lambda_{platform} \cdot Headway$$

Equivalently, using the arrival time:

$$N_{platform} = \frac{Headway}{T_{platform}}$$

Thus, the formal definition of the metric is:

$$N_{platform} = \lambda_{platform} \cdot Headway = \frac{Headway}{T_{platform}} \quad (13)$$

Where $N_{platform}$ is the expected number of passengers accumulated on the platform, $\lambda_{platform}$ is the passenger arrival rate (passengers per minute), and *Headway* is the train Headway (minutes).

Table 4 shows the arrival times for each system platform. The passenger arrival rates reported in this table were obtained from METRO REC ridership records for the modeled stations, expressed as the average number of passengers arriving per minute at each platform, from which the corresponding mean inter-arrival times were derived. Labels referring to arrival times on each platform were defined and associated with the performance submodel. For platform LAR1, for instance, the mean number of passengers waiting for the arrival of the train to board is given by:

$$N^{\circ}Pass_{LAR1} = \left(\frac{Headway}{Arrival_{LAR1}}\right) \quad (14)$$

For the remaining station platforms, the corresponding expressions can be obtained by performing the appropriate substitutions using the parameter values provided in Table 4.

Number of passengers per train

Calculates the number of passengers who boarded the trains in operation. Note that the previous metric defines that passengers arrive at a certain rate during a time that corresponds to the train interval (Headway).

This is because, as soon as the train arrives, it occupies the track circuit corresponding to the platform where the passengers arrived, allowing all passengers who arrived up to that moment to board the train.

Thus, at each Headway, the trains in operation "collect" the passengers who arrive on the system's platforms. In other words, to know how many passengers boarded 1 train ($N^{\circ}(passengers/train)$), simply add the number of passengers per platform of all the track circuits in the system. Thus, we have the following.

$$N^{\circ}(passengers/train) = \left[\sum_{i}^{n}\left(\frac{Headway}{Arrival_{platform}}\right)(i)\right] \quad (15)$$

Table 4
Data on passenger arrivals at station platforms.

Platform	Rate	Arrival (min)	Label
LAR1	6	0.167	Arrival_LAR1
LAR2	3.5	0.286	Arrival_LAR2
GUA1	10	0.1	Arrival_GUA1
GUA2	5	0.2	Arrival_GUA2
PRZ1	35	0.029	Arrival_PRZ1
PRZ2	12.5	0.08	Arrival_PRZ2
CAJ1	52.5	0.019	Arrival_CAJ1
CAJ2	22	0.045	Arrival_CAJ2

5.3. Performability model

The proposed performability model integrates the RTSCS device availability submodels with the railway performance submodel. Consequently, this unified model enables the computation and joint analysis of the performance and availability metrics presented in the previous subsections. In addition, the performability model supports variations in the number of trains in operation, a scenario further examined in Section 7.1, as well as the incorporation of new RTSCS devices, which are evaluated in Section 7.2.

This model also enables studies to vary the arrival times of

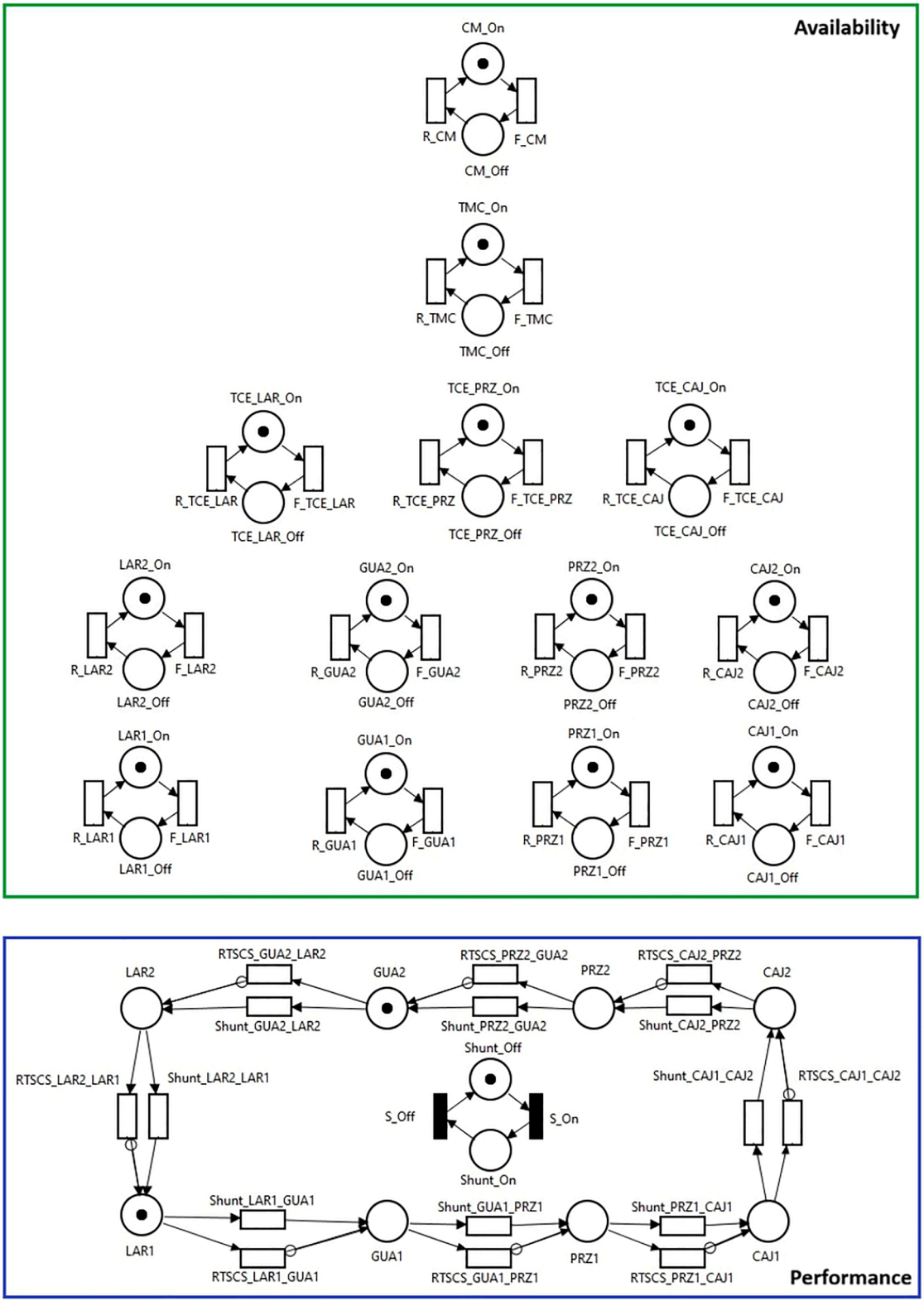


Fig. 8. CAJ-DR performability model.

passengers on platforms (*Arrival* labels previously shown in Table 4). This flexibility is essential for representing scenarios such as football matches or large concerts, as it allows the corresponding increases in passenger arrival rates to be incorporated directly into the model. Fig. 8 presents the proposed performability model, where the availability submodel is highlighted in green and the performance submodel is highlighted in blue.

It is important to clarify that the two submodels do not operate in isolation: they are coupled through the guard expressions assigned to the stochastic transitions that reproduce the movement of the train (token) between the stations (places) of the performance submodel. The third column of Table 3 gathers these guard expressions, which define the firing conditions of each transition. Each guard expression encodes the operational conditions (On or Off, available or unavailable) of the set of devices that represents the command logic for the corresponding track section, considering both the RTSCS and the Shunt modes. Therefore,

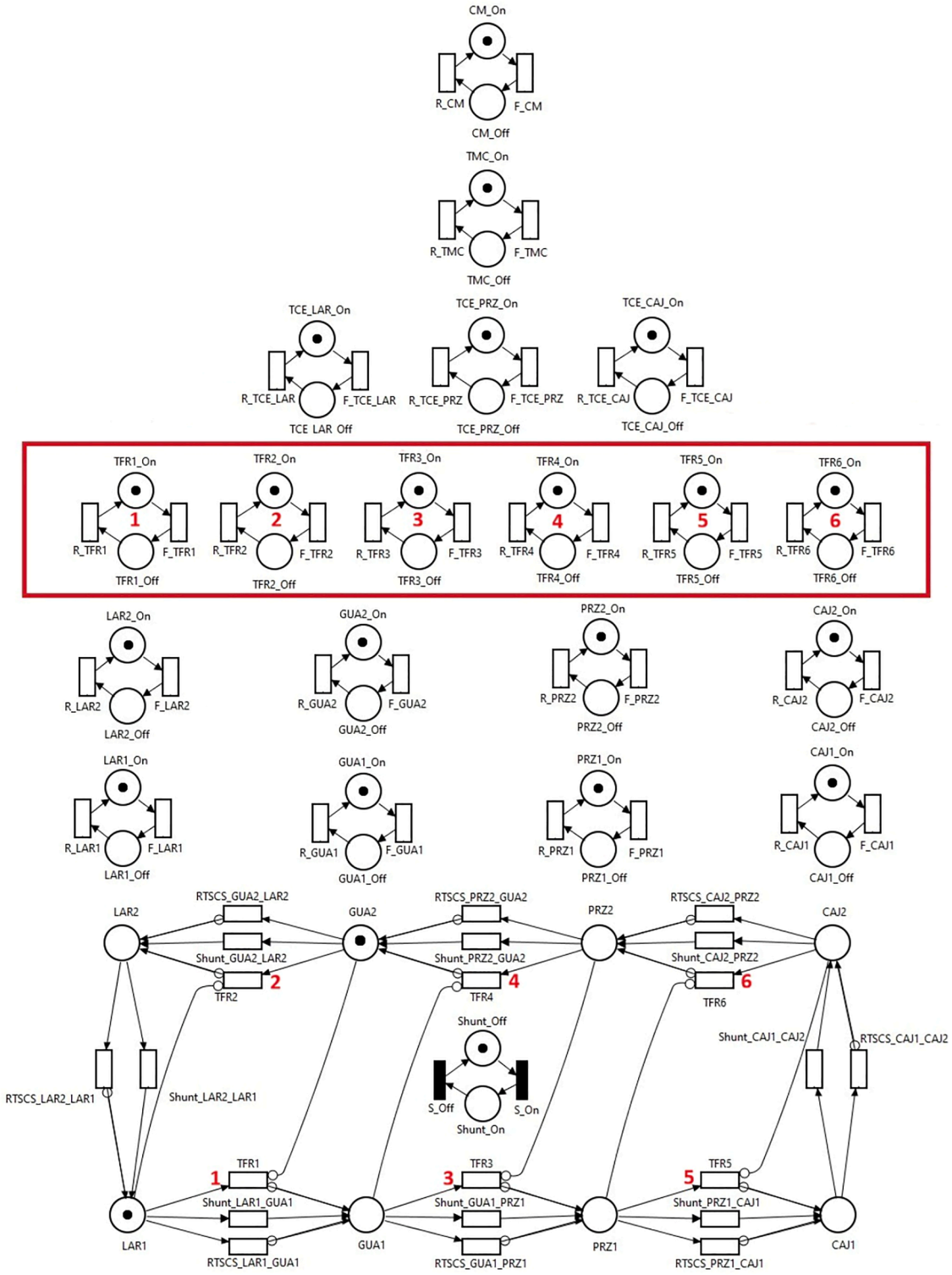


Fig. 9. CAJ-DR performability model with 6 TFRs.

whether a given movement transition in the performance submodel is enabled, and in which mode (RTSCS or Shunt) the train traverses the track circuit, depends directly on the current marking of the availability submodels. In this way, the results of the availability submodels continuously affect the behavior of the performance submodel, which is precisely what integrates both submodels into a single performability model rather than two independent models running in parallel.

5.4. Performability model with TFRs

This section presents a variation of the CAJ-DR performability model in which TFRs is incorporated into selected track circuits to enhance both performance and availability metrics. Fig. 9 shows the performance model with six TFRs added.

Highlighted in red are the availability submodels for each added traffic flow route. For each TFR, an additional submodel representing the failure and recovery behavior of the new route must be included in the availability submodels. Moreover, in the performance submodel, a stochastic transition must be added to the track circuit where the TFR is implemented. The values associated with the failure and recovery transitions ("F_TFR" and "R_TFR") are 55,000 min and 240 min, respectively. These values are consistently applied to the corresponding transitions of all the other added TFRs.

Once the performance and availability of each TFR have been modeled, as illustrated in Fig. 9 (see pairs 1–1 to 6–6), it is necessary to establish the activation criterion (guard expression) for each route. For TFR1, for example, the requirements for activating this route are: 1) failure in the LAR1 track circuit (#LAR1_Off=1); 2) critical devices available (#TCE_LAR_On=1)AND(#TMC_On=1)AND(#CM_On=1); 3) operational traffic flow route (#TFR1_On=1). Thus, the activation of the stochastic transition representing TFR1 can be expressed as: (((#LAR1_Off=1)AND(#TCE_LAR_On=1)AND(#TMC_On=1)AND (#CM_On=1)AND(#TFR1_On=1)). The guard expression that determines the activation of Shunt mode ("S_On") with 6 added TFRs is given by:

(((#CM_Off=1)OR(#TMC_Off=1))OR(((#TCE_LAR_Off=1)AND ((#LAR1=1)OR(#LAR2=1)OR(#GUA1=1)OR(#GUA2=1)))OR ((#TCE_PRZ_Off=1)AND((#PRZ1=1)OR(#PRZ2=1)))OR ((#TCE_CAJ_Off=1)AND((#CAJ1=1)OR(#CAJ2=1))))OR ((((#LAR1=1)AND(#LAR1_Off=1)AND(#TFR1_Off=1))OR ((#LAR1=1)AND(#LAR1_Off=1)AND(#TFR1_On=1)AND (#GUA2>0)))OR((#LAR2=1)AND(#LAR2_Off=1))OR (((#GUA1=1)AND(#GUA1_Off=1)AND(#TFR3_Off=1))OR ((#GUA1=1)AND(#GUA1_Off=1)AND(#TFR3_On=1)AND (#PRZ2>0)))OR(((#GUA2=1)AND(#GUA2_Off=1)AND (#TFR2_Off=1))OR((#GUA2=1)AND(#GUA2_Off=1)AND (#TFR2_On=1)AND(#LAR1>0)))OR(((#PRZ1=1)AND (#PRZ1_Off=1)AND(#TFR5_Off=1))OR((#PRZ1=1)AND (#PRZ1_Off=1)AND(#TFR5_On=1)AND(#CAJ2>0)))OR (((#PRZ2=1)AND(#PRZ2_Off=1)AND(#TFR4_Off=1))OR ((#PRZ2=1)AND(#PRZ2_Off=1)AND(#TFR4_On=1)AND (#GUA1>0)))OR((#CAJ1=1)AND(#CAJ1_Off=1))OR(((#CAJ2=1) AND(#CAJ2_Off=1)AND(#TFR6_Off=1))OR((#CAJ2=1)AND (#CAJ2_Off=1)AND(#TFR6_On=1)AND(#PRZ1>0)))

The guard expression representing the conditions for deactivating Shunt mode (S_Off) is given by the negation of the activation expression (NOT "S_On"). Just as in the model without TFRs, the failure of critical equipment such as the CM or TMC already allows the activation of the Shunt. The failure of a TCE also allows the activation of the Shunt if there is a train occupying a track circuit subordinate to the failed TCE. In the case of failure of a TC occupied by the train, if this track circuit has a TFR, the Shunt mode may be activated in two circumstances: 1) TFR unavailable - shunt mode activated due to TFR failure; 2) TFR available, but with the opposite track occupied - Shunt mode activated due to safety criteria.

For the scenario illustrated in Fig. 9, where trains are occupying track circuits LAR1 and GUA2, even if there is a fault in circuit LAR1 (#LAR1_Off=1), the train cannot activate TFR1 because there is a train occupying the TC on the opposite track (#GUA2=1). However, if GUA2 is not occupied, the shunt can only be activated if the route of that circuit is unavailable (#TFR1_Off=1). Therefore, the guard that expresses this condition is given by:

(((#LAR1=1)AND(#LAR1_Off=1)AND(#TFR1_Off=1))OR ((#LAR1=1)AND(#LAR1_Off=1)AND(#TFR1_On=1)AND (#GUA2>0)))

Note that this expression corresponds to lines 5 and 6 (before the OR) of "S_On". The adjustments for the remaining tripping requirements for the other track circuits with an added flow route follow the same logic adopted in TFR1. The following subsections discuss the necessary adjustments to the availability and performance metrics of models with TFRs.

5.4.1. Availability metrics

The availability metrics consider the addition of a modeled TFR operating in parallel with the track circuit (TC) targeted for optimization. For example, the LAR1 track circuit becomes operational if the LAR1 TC is available (#LAR1_On = 1) or if the traffic flow route added to it is available (#TFR1_On = 1). In addition, all higher-level RTSCS devices required for system operation must also be available; that is, (#TCE_LAR_On=1)AND(#TMC_On=1)AND(#CM_On=1). Thus, the availability of LAR1 track circuit with TFR1 added ($Avail_{LAR1_TFR1}$) becomes:

$$\begin{aligned} Avail_{LAR1_TFR1} &= P\{((\#LAR1_On = 1)OR(\#TFR1_On \\ &= 1))AND((\#TCE_LAR_On = 1)AND(\#TMC_On \\ &= 1)AND(\#CM_On = 1))\} \end{aligned}$$

The same adjustment methodology applied to the LAR1 track circuit is adopted for the other TCs with added TFR. Since the availability of the control chain for each track circuit is considered, the average availability of the system's current transformers (TCs) is given by:

$$Mean_{Availability(TFRs)} = \frac{1}{n}\sum_{i=1}^{n} Availability_{TCs_TFRs} \quad (16)$$

5.4.2. Performance metrics

The performance metrics consider the addition of TFRs (from 1 to 6) in parallel to the TCs, with the exception of LAR2 and CAJ2. Thus, only metrics that are based on a mathematical relationship with the addition of TFRs should undergo adjustments. The number of trains in operation ($N°Trains$), for example, does not need to undergo any adjustment since its expression considers only the sum of the expectations of all places in the performance submodel (Eq. (4)).

Performance metrics account for the addition of TFRs (from 1 to 6) operating in parallel with the track circuits, with the exception of LAR2 and CAJ2. Consequently, only metrics whose mathematical formulations are directly affected by the inclusion of TFRs require modification. For example, the number of trains in operation ($N°Trains$) does not require any adjustment, since its expression depends solely on the sum of the expected values of all places in the performance submodel (Eq. (4)).

The metric that calculates the travel time (Eq. (6)) is based on throughput (mean time that a train/token spends in a TC/place). Thus, once TFR.are added in parallel to the TCs, each track circuit now has three travel options (RTSCS, Shunt, and TFR), which requires an adjustment in the throughput (λ) of each TC that receives a TFR.

Since λ is based on the mean time (Δt_{mean}) that a train (token) spends in a TC (place), this new time base must be adjusted to consider the third travel option that has been made available with the addition of TFR. In this way, the train will now be able to travel, subject to the RTSCS, when the TC occupied by the train is operational or when the corresponding

TFR is available. In other words: "(TC_On=1)OR(TFR_On=1)". In terms of availability, this logical "OR" between the TC and the TFR can be expressed as follows:

$$1 - \left[(1 - Availability_{TC}) \times (1 - Availability_{TFR})\right]$$

Following the same methodology adopted in Eq. (7), which calculates the average train travel time based on the LAR1 track circuit (Δt_{LAR1_TFR1}), and making the necessary adjustments considering the new availability, the average travel time (based on LAR1 with an added TFR) becomes expressed as:

$$\begin{aligned}\Delta t_{\mathrm{LAR1_TFR1}} = &[(1-(1-\mathrm{P}\{(\#\mathrm{LAR1_On}=1)\mathrm{AND}(\#\mathrm{TCE_LAR_On}=1)\mathrm{AND}(\#\mathrm{TMC_On}=1)\mathrm{AND}(\#\mathrm{CM_On}=1)\}))\\ &\times(1-\mathrm{P}\{(\#\mathrm{TFR1_On}=1)\mathrm{AND}(\#\mathrm{TCE_LAR_On}=1)\mathrm{AND}(\#\mathrm{TMC_On}=1)\mathrm{AND}(\#\mathrm{CM_On}=1)\})\times\Delta \mathrm{t}_{\mathrm{RTSCS_LAR1}}]\\ &+[(1-(1-\mathrm{P}\{(\#\mathrm{LAR1_On}=1)\mathrm{AND}(\#\mathrm{TCE_LAR_On}=1)\mathrm{AND}(\#\mathrm{TMC_On}=1)\\ &\mathrm{AND}(\#\mathrm{CM_On}=1)\}))\times(1-\mathrm{P}\{(\#\mathrm{TFR1_On}=1)\mathrm{AND}(\#\mathrm{TCE_LAR_On}=1)\mathrm{AND}\\ &(\#\mathrm{TMC_On}=1)\mathrm{AND}(\#\mathrm{CM_On}=1)\}\times\Delta t_{\mathrm{Shunt_LAR1}}].\end{aligned} \tag{17}$$

Applying the same procedure to all track circuits (with the exception of LAR2 and CAJ2, which do not have an added TFR) and adopting the same formalism used in Eq. (18), we obtain:

$$MeanTravel_time(TFRs) = \left[\sum_{i}^{n}\left(\frac{N^{\circ}Trains}{\lambda_{TCs_TFRs}}\right)(i) \Big/ n\right] \tag{18}$$

Once the mean travel time (*MeanTravel_time*) has been determined, and considering that the Headway (the mean time interval between trains) is derived from this metric, it is sufficient to apply the corresponding adjustments to obtain the metric that expresses the new Headway ($Headway_{TFRs}$). Thus, we have:

$$Headway_{TFRs} = \left(\frac{MeanTravel_time(TFRs)}{N^{\circ}Trains}\right) \tag{19}$$

Continuing with the adjustments, the metric that computes the value of the number of trips per hour (*N°Trips/hour*) will consider the total number of trips (after adding the TFRs) made by trains in operation within a 1-hour (60-minute) interval. Thus, we have the following.

$$N^{\circ}Trips \Big/ hour = \left[\left(\frac{60}{MeanTravel_time(TFRs)}\right) \times N^{\circ}Trains\right] \tag{20}$$

The metric that expresses the probability of dual TC occupancy does not require any adjustments and remains as defined in Metric 12 (*Prob. DualTC*).

Finally, the metrics that quantify the number of passengers per platform and the number of passengers per train must be updated, since both depend on the Headway. The expressions are given by the following metrics.

$$N^{\circ}Pass_LAR1 = \left(\frac{Headway_{TFRs}}{Arrival_{LAR1}}\right)(1) \tag{21}$$

In conclusion of the adjustment of the performance metrics, we have the number of passengers per train, which will be:

$$N^{\circ}_{(Passengers/train)} = \left[\sum_{i=1}^{n}\left(\frac{Headway_{TFRs}}{Arrival_{TCs}}\right)(i)\right] \tag{22}$$

6. Validation

This section deals with the mathematical validation process of the proposed models. To this end, this research considered a reference scenario (with a train in operation) to be used as a standard for comparison between the expected results (calculated mathematically from data provided by the system operator) and the results obtained by the steady-state simulation of the proposed SPN models.

It is worth highlighting that this scenario (1 train in operation) was selected because it is not impacted by the actions of another train in operation in the system (Queueing Theory, Little law). Therefore, once a mathematical confidence interval (CI) of 1% is established, the expected results can be compared with the simulated results. If these are within the established CI, the model is considered valid. Once validated, the variation of trains in operation in the system can be simulated.

It is important to clarify that the use of real data and the mathematical computation are not competing validation arguments, but complementary roles within the same process. Real operational data are used to feed the availability submodels with the MTTF and MTTR values, obtained from the equipment failure records, which are required to compute the availabilities of the signaling and control devices. The scenario with one train in operation is adopted as the reference precisely because it is not affected by the presence of other circulating trains (Queueing Theory), so that closed-form analytical expressions can be derived for the evaluated metrics. The actual validation argument remains the comparison between the results of the stationary simulation of the SPN models and the values obtained by the mathematical calculation, both evaluated for the single-train scenario (since the impact of other trains is not considered). Once the simulated and the mathematically calculated results lie within the established confidence interval, the validation step of the model and its metrics is regarded as fulfilled. In other words, the real data provide realistic inputs to the models, whereas the agreement between simulation and mathematical calculation provides the validation itself.

Regarding the choice of this value, the stationary simulations were performed in the adopted tool using a confidence level of 95% and, by default, a maximum relative error of 10%. In this work, however, a stricter acceptance criterion was deliberately adopted: rather than relying on the default maximum relative error, the validation required the simulated values to differ from the analytically computed ones by at most 1%. This 1% bound is therefore not a property of the tool but an acceptance threshold defined in this research to make the validation as tight as possible, allowing only a very small deviation between the simulated and the analytical results and thus providing stronger

Table 5
Availability of RTSCS devices.

Device	Availability
CM	0.999
TMC	0.998
TCE_LAR	0.997
TCE_PRZ	0.996
TCE_CAJ	0.997
TFRs	0.996
LAR1	0.859
LAR2	0.863
GUA1	0.862
GUA2	0.864
PRZ1	0.860
PRZ2	0.861
CAJ1	0.864
CAJ2	0.858

Table 6
Result of the availability metrics of TCs.

Track circuit	Metric	Result
LAR1	$Avail_{LAR1}$	0.854
LAR2	$Avail_{LAR2}$	0.858
GUA1	$Avail_{GUA1}$	0.857
GUA2	$Avail_{GUA2}$	0.859
PRZ1	$Avail_{PRZ1}$	0.855
PRZ2	$Avail_{PRZ2}$	0.855
CAJ1	$Avail_{CAJ1}$	0.858
CAJ2	$Avail_{CAJ2}$	0.853
Mean TCs	$(1/n)\ \Sigma_{i=1}^{n}\ Avail_{TCs}(i)$	0.856

Table 7
Data from CAJ-DR track circuits.

Track circuit	$Mean_{speed}$ (m/s)	Δs (meters)
LAR1	11.667	872.73
LAR2	2.778	200
GUA1	11.944	1039.07
GUA2	11.667	872.73
PRZ1	12.222	1138.51
PRZ2	12.222	1039.07
CAJ1	2.778	200
CAJ2	13.056	1138.51

evidence that the SPN models reproduce the expected behavior.

Thus, starting with the availability submodel (Fig. 6), the availability of each RTSCS device can be obtained through the mean time to failure and repair (MTTF and MTTR - Table 2). It is worth noting that the MTTF and MTTR of the TFRs are not listed in this table; these values are 55,000 and 240 min, respectively.

The availability of a computer system can be defined as the fraction of time the system is operational relative to the total time considered. In terms of reliability and maintenance measures, this metric is given by the ratio between the MTTF and the sum of the MTTF and MTTR [23].

$$Availability = \frac{MTTF}{(MTTF + MTTR)} \tag{23}$$

Table 5 shows the availability of each RTSCS device. Note that the availability of TFRs (line 6) expresses exactly the result of 55,000/(55,000 + 240) ≈ 0.996. Since the availability of each track circuit assumes that all devices above each TC must be operational (see Metric 2 for track circuit LAR1), we have:

$$\begin{aligned} Avail_{LAR1} &= P\{(\#LAR1_On = 1)AND(\#TCE_LAR_On \\ &= 1)AND(\#TMC_On = 1)AND(\#CM_On = 1)\} \\ &= 0.859 \times 0.997 \times 0.998 \times 0.999 \approx 0.854 \end{aligned}$$

The same methodology applies to the other track circuits of the evaluated system. Table 6 shows the availability of all the TCs involved. The same methodology applies to the other track circuits in the evaluated system. The table shows the availability of all the TCs involved. Thus, the result of the average availability of the track circuits involved in the evaluated system can be seen in the last line of Table 6 ($Avail_{TCs}$ = 0.856 = 85.6%).

Once the average availability of the system's TCs has been calculated, the next step is to calculate the mean travel time of the trains.

It is important to emphasize that the average travel time (Metric 9) is the base metric for all other metrics in the performance submodel (Fig. 7). Table 7 compiles real data provided by the Permanent Way Coordination Office of the company operating the evaluated railway system [9]. This Table shows the distance (Δs) data between the track circuits of each station, as well as the train traffic speeds ($Mean_{speed}$) in each section.

Using the data presented in Table 7 and being aware of the classical equation that determines the mean speed ($Mean_{speed}$), where $Mean_{speed}$ corresponds to the fraction of the distance traveled by the time interval spent to travel that distance (Δt). For track circuit LAR1, for example, the time that the train spends traveling the distance of this track circuit, when traveling at the mean operational speed established by the RTSCS, can be obtained by the following calculation:

$$\begin{aligned} Mean_{speed} &= \frac{\Delta s}{\Delta t} \Rightarrow \Delta t = \frac{\Delta s}{Mean_{speed}} \therefore \Delta t_{LAR1_RTSCS} = \frac{\Delta s_{LAR1}}{Mean_{speed}} \\ &= \frac{872.73}{11.667} = 74.803\ s = 1.247\ min \end{aligned}$$

The time in which the train moves in Shunt mode follows the same mathematical method, with the difference that, in Shunt mode, the train moves at a restricted speed of 10 km/h (2.778 m/s) for security reasons. Thus, the time interval in the Shunt mode for the LAR1 track circuit (Δt_{LAR1_Shunt}) is:

$$\Delta t_{LAR1_Shunt} = \frac{\Delta s_{LAR1}}{Restricted_{speed}} = \frac{872.73}{2.778} = 314.158\ s = 5.236\ min$$

With the time intervals for each mode (RTSCS and Shunt) and the result of the calculation of the availability of the track circuit, it is possible to calculate the time interval in which the train will consider the availability of the TC associated with the travel time in RTSCS mode, added to the time interval in which the train considers the unavailability of the TC associated with the travel time in Shunt mode ($\Delta t_{(RTSCS\&Shunt)}$). Thus, for TC LAR1, we have:

$$\begin{aligned} \Delta t_{LAR1(RTSCS\&Shunt)} &= (Avail_{LAR1} \times \Delta t_{LAR1_RTSCS}) \\ &\quad + ((1 - Avail_{LAR1}) \times \Delta t_{LAR1_Shunt}) \\ &= (0.854 \times 1.247) + ((1 - 0.854) \times 5.236) \\ &\approx 1.829\ minutes. \end{aligned}$$

Table 8 presents the values of the time intervals for each train movement mode on all track circuits. Note that the first line of Table 8 shows the results calculated for TC LAR1. The same procedure adopted to obtain the time values in LAR1 applies to the other track circuits in the system.

Once the time intervals ($\Delta t_{(RTSCS\&Shunt)}$) of all track circuits are obtained, the travel time that the train takes to travel the evaluated section of the system can be calculated.

$$\begin{aligned} TravelTime &= 1.829 + 1.2 + 2.134 + 1.81 + 2.318 + 2.116 \\ &\quad + 1.2 + 2.244 \\ &= 14.851\ minutes \end{aligned}$$

The Headway, as already presented by the Metric 10, is the result of the fraction of *TravelTime* by the number of trains in operation ($N^{\circ}Trains$). Since the selected scenario has only one train traveling on the railway, the Headway will be:

$$Headway = \frac{TravelTime}{N^{\circ}Trains} = \frac{14.851}{1} = 14.851\ min$$

Once the travel time of a train has been calculated, and considering what the Metric 11 ($N^{\circ}Trips/hour$) expresses, the number of trips per hour will be:

Table 8
Values of TCs time intervals for each train travel mode.

Track Circuit	Δt_{RTSCS} (min)	Δt_{Shunt} (min)	$\Delta t_{(RTSCS+Shunt)}$ (min)
LAR1	1.247	5.236	1.829
LAR2	1.2	1.2	1.2
GUA1	1.45	6.234	2.134
GUA2	1.247	5.236	1.81
PRZ1	1.553	6.831	2.318
PRZ2	1.417	6.234	2.116
CAJ1	1.2	1.2	1.2
CAJ2	1.453	6.831	2.244

$$N^{\circ}Trips \Big/ hour = \left(\frac{60}{TravelTime}\right) \times N^{\circ}Trains = \frac{60}{14.851} \times 1 = 4.04\ trips \Big/ hr$$

Since the scenario selected for the validation of the metrics involves only one train in operation, the probability of dual TC occupancy is zero. The possibility of a dual TC occupancy arises only in scenarios with two trains operating simultaneously. In this case, with two trains traveling on the railway, the system can be analyzed using Little's queueing theory, as the introduction of an additional train may affect the travel time. Therefore, for Metric 12, a mathematical calculation is not required.

The number of passengers per platform is determined by considering the Headway and the mean passenger arrival time at each platform associated with the track circuits. Accordingly, based on the definitions provided by Metric 14 ($N^{\circ}Pass_{LAR1}$) and the mean passenger arrival times reported in Table 4, the following results are obtained.

$$N^{\circ}Pass_{LAR1} = \frac{Headway}{Arrival_{LAR1}} = \frac{14.851}{0.167} = 88.928\ passengers$$

Once the updated number of passengers per platform has been obtained for each track circuit, it is possible to compute the corresponding number of passengers per train as follows:

$$N^{\circ}pass \Big/ train = \sum_{i=1}^{n} \left(\frac{Headway}{Arrival_{TCs}}\right)_i = 88.928 + 51.927 + 148.51 + 74.255 + 512.103 + 185.638 + 781.632 + 330.022 = 2173.015\ passengers \Big/ train$$

Based on the results of all defined metrics, it is possible to compare the values obtained through analytical calculations with those derived from stationary simulations performed using the Mercury modeling tool. Table 9 presents the results obtained by both approaches for the scenario involving a single train operating in the SPN performability model.

It should be noted that a mathematical confidence interval, denoted as C.I, of 1 percent was established. Consequently, the metric values obtained by stationary simulation are expected to fall within this C.I, corresponding to a variation of plus or minus 1 percent relative to the values obtained by mathematical calculation (Math).

It is important to clarify that the validation was not restricted to the mean travel time. The metrics presented in Table 9 were selected as reference metrics for the validation, and they cover the different dimensions of the proposed performability model: the mean availability of the track circuits, $Mean_{avail(TCs)}$, which is governed by the failure and recovery (MTTF and MTTR) parameters of the availability submodels; the mean travel time; the Headway; the number of trips per hour; and the passenger accumulation per train. In every case, the value obtained by stationary simulation falls within the established 1% confidence interval relative to the value obtained by the mathematical calculation, so that all of these metrics, and not only the travel time, are validated. The same reasoning extends to the remaining metrics of the model: because they are computed from the same validated SPN structure and parameters and are likewise compared against their analytical expressions, they also fall within the established confidence interval and are therefore equally validated. The availability metric in particular confirms the validation of the failure and recovery behavior of the equipment, since $Mean_{avail(TCs)}$ is a direct function of the MTTF and MTTR values discussed previously. With respect to the probability of dual TC occupancy, as already noted, in the reference scenario adopted for validation (one train in operation) this probability is necessarily zero, since at least two trains on the same track circuit are required for a dual occupancy to be possible; therefore, this metric does not admit a mathematical comparison in the validation scenario and is instead evaluated in the case studies, where more than one train is in operation.

Table 9
Comparison of the results obtained via mathematical calculation with those obtained via stationary simulation.

Metric Evaluated	Results (Math)	[$Value_{Min}$; $Value_{Max}$] (C.I)	Results (Simulation)
$Mean_{avail(TCs)}$	0.856	[0.848; 0.865]	0.856
$Mean_{Travel_time}$	14.851	[14.702; 14.999]	14.744
Headway	14.851	[14.702; 14.999]	14.744
$N^{\circ}_{(Trips/hour)}$	4.040	[4.000; 4.080]	4.070
$N^{\circ}_{(pass/train)}$	2175.652	[2153.896; 2197.409]	2157.301

7. Case studies and results

This section presents two structured case studies to illustrate the applicability of the proposed models. In the first case study, the research examined the variation in the number of trains operating on the railway (from 1 to 8 trains). The main objective of the first case study is to identify the optimal number of trains in operation based on the evaluated performance and availability metrics. In the second case study, traffic flow routes were added to the performability model (ranging from 1 to 6 TFRs). The focus of this new case study is to increase the availability of the system in a way that positively impacts the evaluated performance and availability metrics.

It is important to note that the performability model without the addition of TFRs represents the normal condition of the railway, while the performability model with TFRs added to the track circuits (TCs) represents the system improvement proposal presented in this research. The results of the evaluation of the metrics of the proposed performability model are described in the subsection corresponding to each case studied.

7.1. Case study I

Fig. 10 illustrates a graph that shows the impact of the variation in the number of trains on performance indicators such as the mean travel time of a train ($Mean_{Travel_time}$ - blue bar) and the Headway between trains (Headway - orange bar). This graph shows that $Mean_{Travel_time}$ starts at 14.744 min (1 train in operation) and increases at a mean rate of approximately 0.47% per additional train in operation, reaching a maximum value of 15.227 min in a scenario with eight trains in operation.

The Headway, on the other hand, showed a reduction of approximately 50.0% when comparing the scenarios with 1 and 2 trains operating (from 14.744 min to 7.372 min). With three trains in operation, the Headway drops to 4.915 min (a reduction of about 66.7% compared to the scenario with 1 train in operation). The Headway continues to decrease as more trains are added, reaching a minimum value of 1.903 min with eight trains in operation. The dashed line, fitted by a power model, reflects the Headway trend as a function of the number of trains in operation. The coefficient of determination (R^2) shown in the graph quantifies how well the model explains the observed variance in Headway values; values of R^2 close to 1 indicate that a large proportion of the variability in Headway is captured by the model, reflecting a strong relationship between the number of trains in operation and Headway [24].

Fig. 11 illustrates the impact of the number of trains in operation on the mean availability of the track circuits (TCs Availability – blue line), the probability of dual TC occupancy (Dual TC Occupancy Probability – red line), and the number of trips performed per hour (Trips per Hour – green line). As expected, the curve corresponding to TCs Availability remains practically stable as the number of trains increases. This behavior occurs because the number of trains circulating in the system is not directly related to the intrinsic reliability parameters of the track

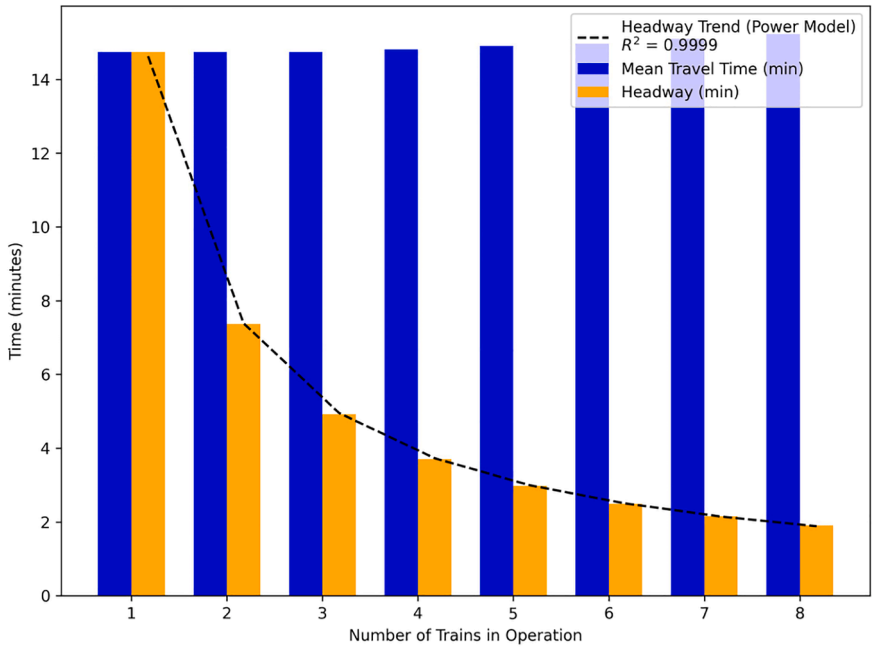


Fig. 10. CAJ-DR without TFRs and variation in the number of trains in operation.

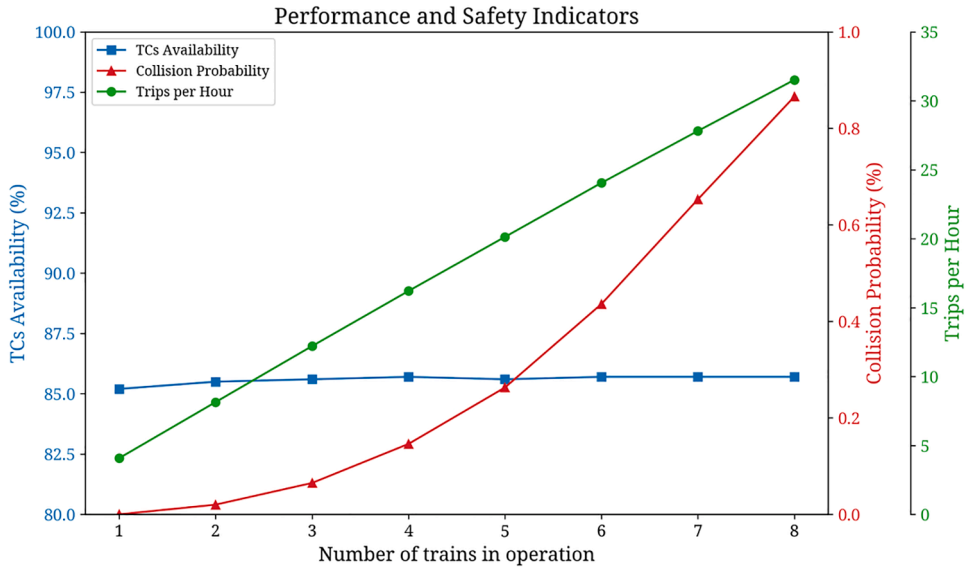


Fig. 11. Number of trains in operation versus mean TC availability, probability of dual TC occupancy and number of trips made per hour.

circuits. Quantitatively, considering the eight evaluated scenarios (from 1 to 8 trains in operation), the mean TCs availability is 0.8559 (85.59%), with a standard deviation of only 0.00,173 and a coefficient of variation of 0.20%. Such a low coefficient of variation confirms the negligible dispersion of availability values across the scenarios, reinforcing the independence between the number of trains in operation and the availability of the track circuits.

The probability of dual TC occupancy starts at zero (1 train in operation, a scenario where there is no possibility of a dual occupancy), increases to 0.020% (2 trains in operation), and continues to increase, reaching 0.146% with 4 trains in operation and a maximum value of 0.866% with 8 trains operating. It is worth recalling that this metric does not consider the driver's reaction to avoid a possible collision between trains (for example, activating the emergency brake). In Fig. 11, it can still be observed that with one train operating, the number of trips made per hour is 4.070. When one more train is added (2 trains in operation), the number of trips made per hour becomes 8.140 (an increase of 100% compared to the previous scenario). From the third train onward, the impact of adding one more train in relation to the previous scenario corresponds to an average increase of 25.90%. The number of trips per hour reaches its maximum value of 31.523 trips/h in the scenario with eight trains operating on the railway.

It is important to clarify the meaning of this dual-TC-occupancy value, which should not be interpreted as the rate at which collisions actually occur in the real system. The metric quantifies the probability of dual TC occupancy in a deliberately conservative (worst-case) scenario in which the train is not subordinated to the signaling and control system, that is, the Shunt mode. In this mode, as described in Section 5.2, the train is not governed by the RTSCS, the operational control center loses the ability to locate the train on the faulty track circuit, and the ATC and the associated safety criteria are inhibited, so the train can advance to an already-occupied track circuit. The reported probability

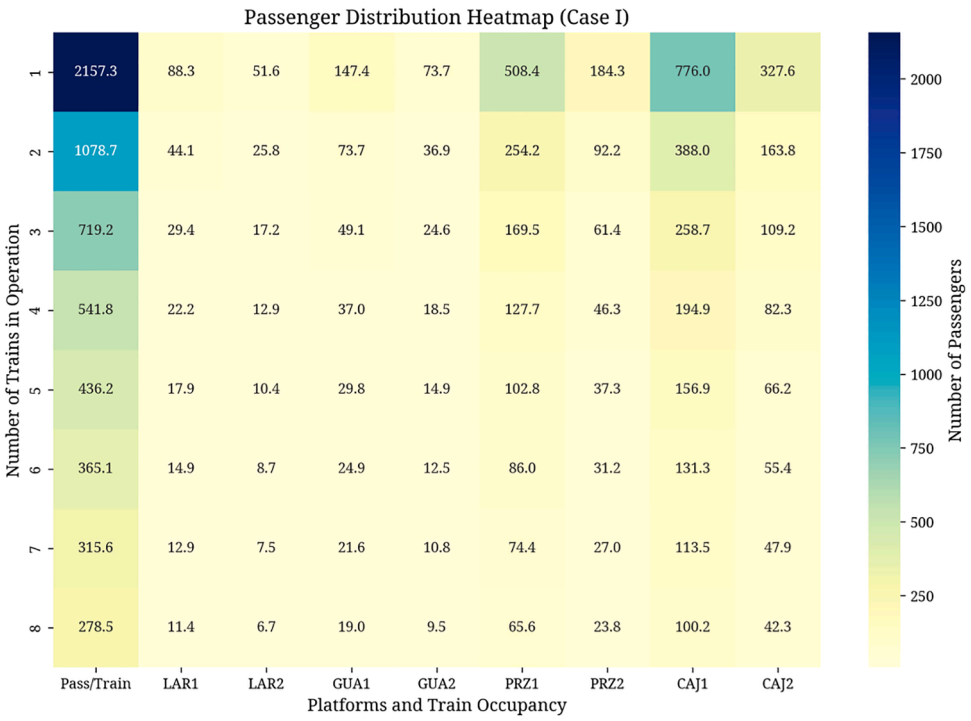


Fig. 12. Heatmap of the number of passengers per train and per platform at each CAJ-DR station in relation to the number of trains operating in the system.

therefore corresponds to the situation in which the driver does not apply the emergency brake, whether due to an unforeseen event, an incident, or a lapse of attention, and is thus left solely to the driver's vigilance to avoid the collision.

The purpose of the metric is precisely to quantify this residual risk, that is, the exposure to collision that remains when the automated protection provided by the RTSCS is unavailable and the only remaining barrier is the human reaction. In normal operation, when trains travel subordinated to the RTSCS, this protection layer is active and the automatic train control prevents such occurrences; the high value reported here reflects the absence of that protection in the Shunt scenario rather than the frequency of real collisions on the line. This conservative formulation is intentional, as it allows the model to measure how much the infrastructure redundancy mechanisms (TFRs) reduce the exposure to collision risk, as quantified in Case Study II. For this reason, the reported probability of dual TC occupancy should be understood as a deliberate over-estimation, that is, a conservative upper bound rather than a realistic operational figure: by removing the automated protection of the RTSCS and disregarding the driver's reaction, the model concentrates in this single metric the worst-case exposure to collision, which explains why its value appears high and, in absolute terms, non-acceptable for normal operation. Under nominal conditions, with the RTSCS and the ATC active, the actual exposure is substantially lower than the value reported for the Shunt scenario.

Fig. 12 shows the Heatmap of the number of passengers per platform and per train in relation to the number of trains operating on the rail. The graph shows that the darker colors refer to higher values of the number of passengers waiting for the train per platform and the number of passengers transported by train.

The Heatmap analysis reveals critical capacity constraints: with one train operating, passenger load reaches 2157 passengers/train (exceeding comfortable capacity of 1000). Platform saturation is evident at CAJ1 (775.9 passengers vs. 250-passenger capacity). With three trains in operation, passenger load reduces to 719 passengers/train, meeting service quality criteria and serving as the baseline for Case Study II. These results demonstrate that insufficient fleet size generates structurally unstable service conditions characterized by excessive crowding, degraded service quality, and reduced travel satisfaction. From an urban systems perspective, such congestion discourages public transport use and undermines sustainability objectives. Case Study I establishes the direct link between infrastructure availability, fleet size, passenger experience, and metropolitan mobility resilience.

It should be emphasized that these crowding scenarios are intentionally modeled to reproduce the saturation of the system, that is, configurations in which few trains are circulating while the passenger arrival flow at the station platforms is maintained. The purpose is to illustrate operating conditions that should be avoided by the operational control center and by the system managers: when the number of passengers waiting to board approaches a value set as a crowding alert threshold, the manager can inject additional trains into the system to handle the demand. The crowding metric therefore serves as a decision-support indicator for fleet dispatching rather than as a description of a normal operating condition, and its safety-critical implications reinforce the importance of keeping passenger accumulation below the alert threshold. In particular, the reference value can be set so as to anticipate conditions that increase the risk of a passenger falling from the platform onto the track, a recognized railway safety hazard associated with platform overcrowding; when the simulated number of passengers per platform approaches this alert value, the operational control center can inject additional trains into the railway to relieve the accumulated demand at the platforms.

Taken together, these results allow a correlated risk analysis, in which the risks captured by the model are not treated in isolation but as interdependent consequences of the same underlying conditions. The availability of the signaling and control devices is the common driver: when these devices become unavailable, the train operates in Shunt mode, which simultaneously increases the exposure to collision risk (since the automated protection of the RTSCS is absent) and degrades the performance metrics (longer travel times and headways), reducing the system's capacity to relieve the platforms. This degradation, in turn, is correlated with the crowding risk, because fewer effective train movements increase passenger accumulation and therefore the risk of a passenger falling from the platform onto the track. Conversely, the addition of Traffic Flow Routes (TFRs), evaluated in Case Study II, acts

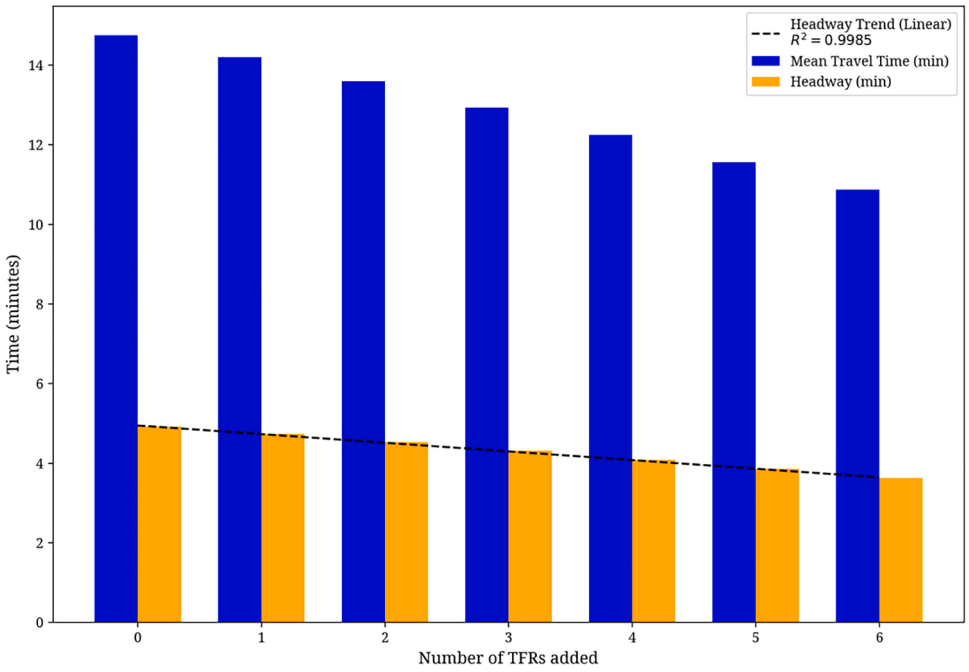


Fig. 13. Impact of adding TFRs on mean train travel time and Headway.

on this same chain by restoring availability, which jointly reduces the collision exposure and the crowding levels. In this sense, the proposed performability model links the availability of the signaling and control infrastructure to both the safety-related risk (collision) and the service-related risk (crowding and platform-fall exposure), providing an integrated view of these correlated risks rather than a set of independent indicators.

7.2. Case study II

The second case study evaluates the addition of the number of TFRs to the proposed performability model. Taking as a basis the scenario defined in Case Study I (3 trains in operation), the TFRs are added to the model, and the performance and availability metrics are evaluated shortly thereafter. Thus, the study conducted in this case varied the addition of 1 to 6 TFRs to the system and evaluated the impact of adding these TFRs on the performance and availability metrics of the proposed performability model.

Fig. 13 illustrates the impact of adding TFRs on the mean train travel time ($Mean_{Travel_time}$ - blue bar) and on the mean Headway time between trains (Headway - orange bar). The graph shows that the $Mean_{Travel_time}$ starts at 14.745 min (scenario without TFRs), decreases to 14.200 min when a TFR is added, and continues to decrease at a mean rate of approximately 4.95% per TFR added, reaching its minimum value of

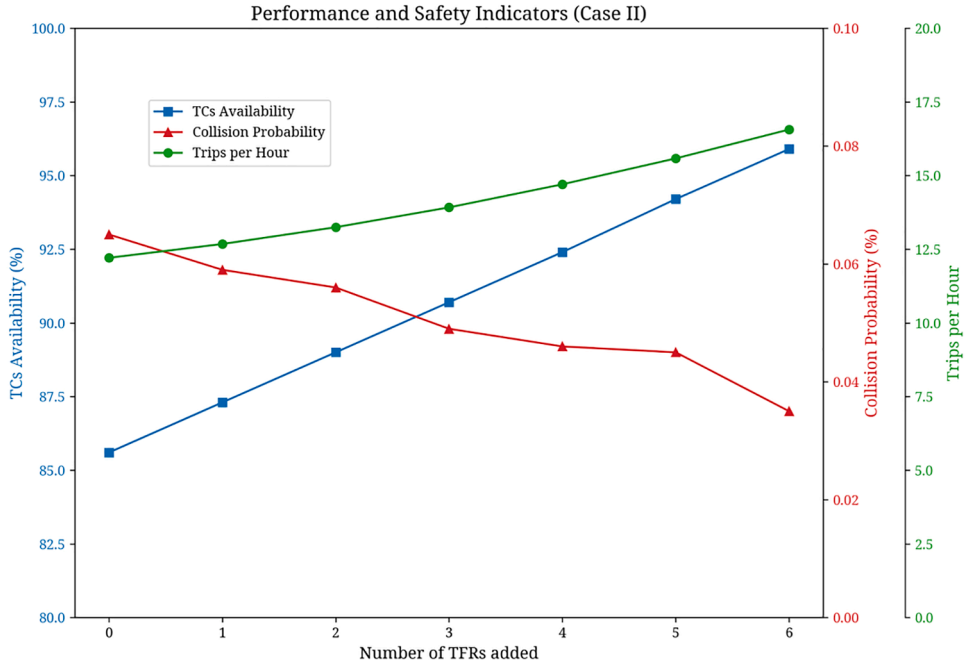


Fig. 14. Impact of adding TFRs on the TCs Availability, Dual TC Occupancy Probability and Trips per Hour.



Fig. 15. Heatmap of passenger accumulation per platform and passenger load per train as a function of the number of TFRs. Columns represent TFR count (0 to 6); rows represent platforms and the train. Increasing TFR count consistently reduces both platform congestion and train load, with the most pronounced reduction observed at critical platforms such as CAJ1 (−26.3%).

10.868 min when 6 TFRs are added to the CAJ-DR performability model.

The Headway starts at 4.915 min (without TFRs) and decreases at a mean rate of approximately 4.95% for each TFR added to the system. The Headway reaches its minimum value of 3.623 min when 6 TFRs are added to the CAJ-DR performability model. It can also be seen in Fig. 13 that the result of R^2 (associated with the trend line of the Headway) is very close to 1 ($R^2 = 0.9985$), indicating that most of the variability of the dependent variable is explained by the model. In other words, this means that the data points are well adjusted to the estimated trend line, suggesting a strong relationship between the variables analyzed (Headway and number of TFRs).

Fig. 14 shows the impact of adding TFRs on the mean availability of track circuits (TCs Availability - blue line), the probability of dual TC occupancy (Dual TC Occupancy Probability - red line), and the number of trips performed per hour (Trips per hour - green line). In this case, the TCs Availability increases from 0.856 to 0.959, while the probability of dual TC occupancy decreases from 0.065% to 0.035% as TFRs are added. Furthermore, the number of trips per hour shows a significant improvement, rising from 12.207 to 16.562 trips/h.

TCs Availability starts at 0.856 (approximately 85.6%) for no TFRs added to the performance model, increasing at a mean rate of approximately 1.91% per TFR added. TCs Availability reaches 0.907 (90.7%) when 3 TFRs are added and peaks at 0.959 (95.9%) for 6 TFRs added. The probability of dual TC occupancy, which is 0.065% for a scenario with 3 trains operating and no TFRs, drops to 0.059% after the addition of the first TFR and continues to decrease at a mean rate of 9.80% per TFR added, reaching its minimum value of 0.035% of dual TC occupancy when 6 TFRs are added to the system. Without added TFRs, for the scenario with 3 trains in operation, there are 12.207 trips performed per hour. After adding a TFR, this metric increases to 12.676 and continues to increase by approximately 5.22% (on mean) for each TFR added to the performability model. The maximum number of Trips per Hour is reached with 6 TFRs added (16.562 trips per hour).

Fig. 15 illustrates the Heatmap of the number of passengers waiting to board the train by platform and the number of passengers transported by train. It can be seen in the Heatmap that the zero column (no TFRs added) reflects the result of the scenario selected in Case Study I (3 trains in operation). The train, for example, starts with 719.173 passengers per train (0 TFRs) and decreases at a mean rate of approximately 4.96%, reaching its minimum value (530.093 passengers/train) when 6 TFRs are added to the model. The metrics that estimate the number of passengers per platform follow the rate of reduction presented by the train, with this value being 4.96% (on mean) for all platforms.

Also in the Heatmap, it can be seen that the lowest number of passengers per platform (12.667) occurs in LAR2, with 6 TFRs added. For the CAJ1 platform, for example, the addition of 6 TFRs represents a reduction of 26.291% compared to the model without TFRs, going from 258.686 to 190.674 passengers per platform. In conclusion, the results of Case Study II demonstrate that the strategic implementation of TFRs substantially enhances system performability by improving traffic distribution without requiring additional rolling stock. The observed increase in trips per hour and the consistent reduction in passenger load per train indicate a mitigation of crowding conditions, directly affecting perceived service quality and reliability. As previously discussed, reductions in passenger density and headway variability are closely associated with improved commuter experience and lower stress exposure in high-demand rail environments.

From an urban systems perspective, these findings suggest that infrastructure redundancy mechanisms can strengthen railway resilience while simultaneously improving operational stability and safety margins, as evidenced by the decline in the probability of dual TC occupancy. Unlike capacity expansion through fleet growth, TFR implementation represents a structural optimization strategy capable of enhancing service reliability at a comparatively lower scale of infrastructural and environmental cost than large-scale alternatives such as fleet expansion or the construction of new lines. It should be noted, as correctly observed, that a TFR requires the installation of switch machines and the associated signaling system to allow trains to move between tracks, and that switches introduce reliability considerations that differ from those of plain-line track; these elements do entail a cost and were therefore explicitly incorporated into the availability submodels. The benefit of a TFR, however, goes beyond merely adding switches: by

increasing the number of route alternatives in the railway network, it allows specific routes to be configured, such as the reverse route proposed in this work, which is set up in the opposite direction using the parallel track circuit when the original track circuit fails, thereby keeping the train in operation and improving availability.

Therefore, Case Study II illustrates that signaling redundancy is not solely a safety enhancement mechanism but a strategic urban mobility instrument that contributes to congestion mitigation, service attractiveness, and the long-term sustainability of metropolitan transport systems. These results reinforce the central argument of this study: infrastructure availability and redundancy planning are critical levers for improving both technical performance and broader urban outcomes.

8. Conclusion and future works

This study proposed and applied an integrated Stochastic Petri Net (SPN) framework to evaluate how the availability of Rail Traffic Signaling and Control Systems (RTSCS) influences railway performability and broader urban mobility outcomes. By jointly modeling mean travel time, headway, trips per hour, passenger accumulation per platform and per train, infrastructure availability, and the probability of dual TC occupancy, the proposed approach enabled a systemic assessment of how technical reliability propagates into service quality, operational stability, and passenger distribution dynamics.

Case Study I demonstrated that operating with a reduced number of trains leads to structurally unstable service conditions, characterized by excessive passenger accumulation and platform saturation. These findings show that insufficient capacity is not merely an operational inefficiency but a structural driver of degraded reliability and service experience in high-demand urban corridors. The results reinforce the importance of balancing fleet size and infrastructure availability to prevent congestion-induced performance deterioration.

Case Study II extended this analysis by examining the strategic implementation of TFRs as redundancy mechanisms. TCs Availability increased from 0.856 to 0.959 with six TFRs, corresponding to an average gain of approximately 1.91% per additional route. Simultaneously, the probability of dual TC occupancy decreased from 0.065% to 0.035%, representing a mean reduction of approximately 9.80% per TFR. Operational efficiency also improved, with Trips per Hour increasing from 12.207 to 16.562 (mean increase of 5.22% per TFR), while passenger load per train decreased from 719.173 to 530.093 passengers (mean reduction of 4.96%). Critical platforms such as CAJ1 experienced reductions above 26% in passenger accumulation, indicating substantial mitigation of localized congestion.

Beyond these quantitative gains, the results suggest that infrastructure redundancy strategies can enhance railway resilience without necessarily expanding rolling stock. By stabilizing headway and reducing crowding levels, TFR implementation contributes to an improved commuter experience, greater service attractiveness, and strengthened metropolitan mobility performance. In this sense, signaling redundancy emerges not only as a safety enhancement mechanism but also as a strategic instrument for improving the reliability and sustainability of urban transport systems.

The main theoretical contribution of this work lies in integrating availability modeling and passenger-related performance indicators within a unified performability framework. Practically, the findings offer evidence that targeted redundancy planning can simultaneously improve safety, operational efficiency, and congestion management, supporting more resilient and user-oriented transport policies.

Limitations include the case study's specificity to a single metropolitan network; however, the modeling framework remains generalizable. Future research should incorporate additional failure modes, real-time operational data, economic and environmental cost assessments, and artificial intelligence techniques for dynamic TFR activation to enhance system adaptability.

CRediT authorship contribution statement

Osvaldo Marinho dos Santos Neto: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Gustavo Rau de Almeida Callou:** Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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